Predicting Persuasive Comments on r/ChangeMyView

Length, Rhetoric, and Cross-Corpus Validation

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Abstract

Reddit's Change My View (CMV) subreddit offers an observable, comment-level signal of persuasive success: the delta (Δ) award. This study investigates whether delta-award status can be predicted from textual features, and whether explicitly defined rhetorical strategies add predictive information beyond comment length. Using a dataset of 3,971 CMV comments (641 delta-awarded, 16.1%) with thread-grouped train/test splits, an engineered-feature pipeline combining sentiment, tone, rhetorical-appeal, readability, and evidence-marker measures reached held-out PR-AUC 0.3848, below a single-feature word-count baseline (0.5256) and a TF-IDF bag-of-words model (0.5119) — revealing comment length as the dominant predictive signal and motivating the length-controlled analyses that follow. Eight rhetorical strategies (including refutation, justification, and concession), scored on a 1-5 scale by an LLM-based rubric adapted from Labruna et al. (2026), were then added to the engineered baseline on a 2,753-comment scored subset, raising PR-AUC from 0.4434 to 0.5098 (+0.0664), a gain that retains 84% of its value under strict length matching, indicating the gain is not purely a length proxy. Individual-strategy importance depends on the measure used: refutation has the highest SHAP-based model attribution, while justification has the largest univariate rank-biserial effect size. Applying the same approach to Winning Arguments, an independently constructed persuasion corpus, replicates the structural finding that strategy scores add value beyond length, but the strategy-level pattern reverses: justification, not refutation, dominates by both measures. The study's contribution is primarily empirical: a documented account of length's predictive dominance, a length-controlled assessment of rhetorical-strategy value, and a cross-corpus evaluation distinguishing what generalizes from what does not.

1. Introduction

1.1 Background and Motivation

Reddit's Change My View (CMV) subreddit provides a research setting for studying persuasion in naturally occurring online discussions (Priniski & Horne, 2018; Na & DeDeo, 2022). In CMV, an original poster (OP) presents a view they are willing to reconsider, and other participants respond with arguments. When an OP considers a response persuasive, they can award it a *delta* (Δ). This mechanism provides an observable, comment-level indicator of reported persuasive success, rather than a label assigned solely by an external evaluator.

A delta is nevertheless an imperfect outcome measure. It records the OP's reported response within the norms and incentives of the forum; it does not independently establish a lasting change in belief. Deltas may also be withheld for reasons unrelated to the quality of an argument. Accordingly, this study treats delta-award status as an operational label for persuasive success, not as direct evidence of an underlying psychological change.

Predicting delta awards poses a further interpretive challenge. Successful and unsuccessful comments may differ in superficial properties, especially length, as well as in how their arguments are constructed. A model may achieve useful predictive performance by exploiting such surface-level properties without identifying meaningful rhetorical differences. Understanding whether explicitly defined rhetorical strategies add predictive information beyond comment length is therefore central to interpreting what persuasion-prediction models learn.

1.2 Research Problem and Objectives

This project investigates the prediction of delta-award status for CMV comments using textual features. The task is formulated as binary classification at the comment level, with evaluation splits grouped by discussion thread to prevent comments from the same thread appearing in both training and test sets. The analysis examines conventional engineered textual features, comment length, and scores assigned to defined rhetorical strategies.

The investigation evolved in response to an important empirical result: comment length alone outperformed a substantially more elaborate engineered-feature pipeline. This finding sharpened the research question from whether textual properties predict delta awards to **which predictive signals remain after accounting for length**. The subsequent length-controlled comparisons, strategy ablations, and interpretation analyses were developed to address that question. The report describes this progression as it occurred, rather than presenting all later analyses as elements of a fully predetermined design.

The objectives are to characterize the predictive role of textual features and length; assess whether rhetorical-strategy scores provide incremental predictive information; identify which strategies are most informative under different analytical measures; and examine how the findings behave in a second persuasion dataset constructed using a different methodology.

1.3 Research Questions

The project is organized around three research questions:

Table 1. The three research questions, with the Results and Discussion sections that address each.

	Research question	Results	Discussion
RQ1	To what extent can persuasive success be predicted from textual features, and what role does comment length play?	§3.1	§4.1
RQ2	Do rhetorical-strategy measures provide incremental predictive information beyond comment length and other textual features, and which individual strategies are most informative once length is controlled for?	§3.2	§4.1
RQ3	Do these findings hold on an independently constructed dataset, and does the relative importance of individual strategies change?	§3.3	§4.2

RQ1 establishes the predictive setting and the importance of length. RQ2 examines whether the rhetorical-strategy measures add information to the existing feature representation, how their observed contribution changes under length matching, and which individual strategies drive that contribution on the home dataset. RQ3 considers the extent to which the home dataset's findings, including the individual-strategy pattern identified under RQ2, replicate on a second dataset constructed using a different methodology.

1.4 Research Approach and Contributions

The study constructs a dataset of 3,971 comments from CMV discussions (referred to below as the home dataset), including 641 delta-awarded comments (16.1%). It evaluates an engineered textual-feature representation alongside simpler baselines, using thread-grouped train/test separation. The engineered features include measures of sentiment, tone, rhetorical appeals, readability, and evidence markers, extracted from both the reply and the original post, as well as relational features capturing differences and similarities between them. In contrast, the TF-IDF and word-count baselines use comment-only information. Some features were derived using large language models (LLMs).

The initial comparisons reveal that word count alone achieves higher predictive performance than the engineered-feature pipeline on the home dataset's held-out test set. The project therefore examines differences in comment length by outcome and performs length-controlled comparisons. These analyses show that the predictive advantage of several feature representations diminishes markedly when length differences are reduced. They also motivate a more focused investigation of rhetorical features. Length matching reduces an important confound, but it does not by itself establish causal effects or complete independence from length.

Eight rhetorical strategies relevant to CMV discussions, including refutation, justification, concession, and evidence use, are defined and scored using an LLM-based rubric. This approach adapts a general scoring methodology described by Labruna et al. (2026) to a different taxonomy and setting. The resulting scores are evaluated through ablation experiments and examined using both SHAP-based model interpretation and univariate rank-biserial effect sizes. The addition of strategy scores improves predictive performance over the engineered-feature representation on the evaluated subset of the home dataset; a substantial portion of the observed incremental gain also remains in a separately constructed length-matched analysis. The interpretation of individual strategies depends on the metric: refutation has the highest model-based SHAP importance in the home-dataset analysis, whereas justification has the

largest univariate rank-biserial effect size. These measures answer different questions and should not be treated as interchangeable.

The study additionally evaluates the rhetorical-strategy approach on Winning Arguments, a persuasion dataset constructed using a different sampling methodology (Tan et al., 2016). The model architecture and configuration selected on the home dataset were retained, but a new model instance was trained on Winning Arguments' training data and evaluated on its held-out data. This is a transfer of the modeling procedure, not a zero-shot evaluation of the original fitted model. Rhetorical-strategy scores show predictive utility in this second dataset. The relative importance of individual strategies varies across datasets: refutation is the leading rhetorical strategy by SHAP importance in the home dataset, whereas justification leads in Winning Arguments. By contrast, justification has the largest univariate rank-biserial effect size in both datasets. A narrower length-matched check was also performed on Winning Arguments, comparing comment length alone against length combined with the strategy scores, rather than the home dataset's own comparison of a full engineered-feature representation with and without those scores; this narrower check found a broadly similar pattern of retention to the home dataset. However, because it is not directly comparable in scope to the home dataset's own ablation, the external evaluation does **not** establish that the incremental contribution of rhetorical strategies beyond length replicates to the same extent as in the home dataset.

The study's contributions are primarily empirical and analytical: (1) a documented examination of the strength of comment length as a predictor of delta awards and of its implications for model interpretation; (2) a systematic assessment of the incremental predictive value of an eight-strategy rhetorical representation, including length-controlled comparisons and a limited scoring-reliability assessment; and (3) an external evaluation that identifies both recurring predictive utility and differences in strategy-level patterns across datasets. The work adapts existing modeling and scoring approaches rather than claiming a new learning algorithm or establishing causal effects of rhetorical strategies.

1.5 Report Organization

The remainder of the report describes the relevant prior research and the methods used for dataset construction, feature extraction, model training and evaluation, length-controlled analysis, rhetorical-strategy scoring, and external validation. The Results chapter presents the corresponding quantitative findings. The Discussion interprets those findings in relation to the three research questions and prior work, examines limitations of the labels, scoring process, and evaluation design, and outlines directions for future research. The report concludes by summarizing the principal empirical contributions.

References and information on accessing the project's code and data are provided in the final sections.

2. Methods

This chapter describes how the home dataset was constructed, how predictive features were engineered, and how models were trained and evaluated. It then describes how the project's later analyses were carried out: the baseline investigation that motivated a change in research direction, the length-controlled analysis, the rhetorical-strategy scoring and ablation, the strategy interpretation, and the external validation on an independent corpus. Numerical detail is included where it is needed to define a dataset, procedure, or evaluation setting; the corresponding findings and their interpretation are reserved for the Results and Discussion chapters.

2.1 Data Collection and Annotation

Data collection targeted the top 500 all-time posts on r/changemyview, using the Reddit API via PRAW, with the collection approach adapted from the code released by Priniski and Horne (2018) for their own study of the same subreddit. After collection and loading, 425 of the 500 requested posts yielded usable comment-level data, producing 3,974 raw comment rows. Each row corresponds to one reply (comment), not one post or one thread, since the per-comment delta award is the project's unit of prediction.

Three columns anchor each row: the original post's text (`original_post`), the preceding context of the same commenter's own prior turns in the thread (`thread_text`), and the comment's own cleaned text (`final_comment`). The definition of `thread_text` is an explicit design choice: it includes only the same commenter's own earlier turns in the discussion, not the full thread including other participants' replies. This keeps the contextual text consistent in authorial style and sentiment with the comment being scored, at the acknowledged cost of omitting same-side arguments that other commenters may have contributed to the same thread.

A documented data-cleaning pass removed three rows: a delta-awarded comment whose body had been edited post hoc into placeholder text, an automated moderator notice that had been parsed as if it were a participant comment, and a comment whose body consisted of a numeric grid that interfered with a readability measure computed later in the pipeline. This left a final dataset of 3,971 comments, of which 641 were delta-awarded (16.1%) and 3,330 were not (83.9%).

The target label, `is_convincing`, is derived directly from whether a comment received a delta from the original poster. Not every comment in a thread becomes a row, and the two classes are selected differently. A delta-awarded comment becomes a positive row regardless of where it sits in the thread; any of its own descendant replies are discarded. A non-convincing comment becomes a negative row only if it is a *leaf* — it has no replies of its own — in the thread tree; a non-convincing comment that has further replies underneath it does not itself become a row, only its descendant leaves do. No topical filtering or matching was applied to select which leaf comments would serve as negative examples among those eligible. This is a deliberate point of contrast with the external validation corpus described in §2.9, whose negative examples are instead selected to be topically similar to the corresponding positive example.

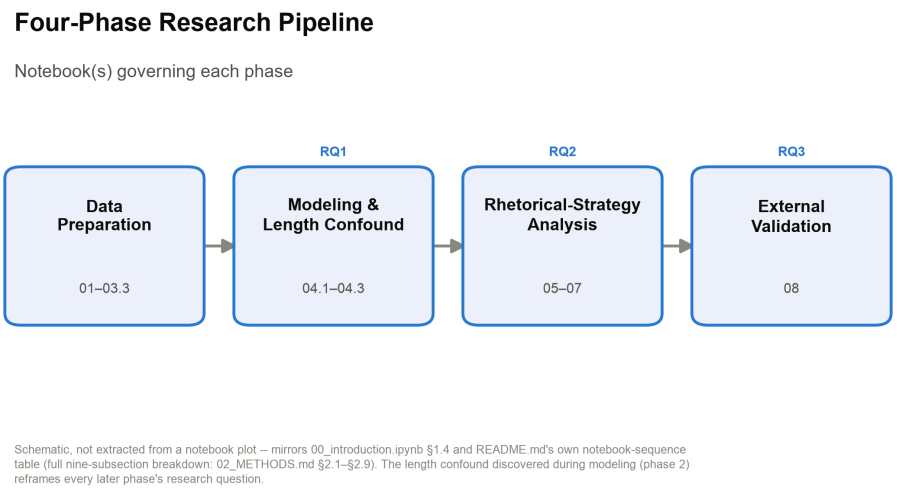


Figure 1. The project's pipeline as four broad phases — data preparation, modeling and the length-confound discovery, rhetorical-strategy analysis, and external validation — each labeled with its governing notebook(s). The last three phases map one-to-one onto the project's three research questions (RQ1–RQ3, tagged on the diagram); data preparation is shared setup, not itself one of the three. The full nine-subsection breakdown is §2.1–§2.9 below.

2.2 Feature Engineering

Feature engineering proceeded in three stages, each implemented in its own notebook: a first pass building linguistic and stylistic features from the comment and the original post; a second pass adding large-language-model-derived features and document embeddings; and a third pass constructing features that relate each comment to its original post directly, instead of describing either side in isolation.

Several feature families went through a documented sequence of trial, diagnosis, and replacement, which is reported here because each rejection reflects a specific, diagnosed shortcoming rather than an unexplained substitution. Sentiment was first estimated with TextBlob, which proved unsuited to comments of the length typical in this dataset, and then with VADER, which was found to conflate a comment's negative *topic* (for example, discussing a negative event) with a negative *tone* toward the original poster's argument; both were superseded by a tone-focused sentiment score assigned by an LLM (Gemini). Tone was first estimated with a RoBERTa-based emotion classifier, which collapsed roughly two-thirds of comments into a single "neutral" category and was judged uninformative as a result; it was replaced by an LLM-assigned categorical tone label with six categories (logical, passionate, humorous, sarcastic, neutral, other), generated for both the comment and the original post. Stylistic register was first estimated with a zero-shot `bart-large-mnli` classifier, which produced visibly incorrect labels on manual spot checks, and was replaced with an LLM-assigned formal/informal label. Rhetorical appeals — ethos, pathos, and logos — were first estimated using the Empath lexicon, which was found to track topic instead of rhetorical structure, and were replaced with LLM-assigned scores on a continuous 0–1 scale for each appeal; of these, the logos score showed the widest separation between convincing and non-convincing comments of any feature introduced at this stage.

Several other features were retained without this trial-and-replacement sequence: an adjective-to-adverb ratio computed with spaCy part-of-speech tagging; two readability measures (Flesch–Kincaid grade level and reading ease) computed with the `textstat` library; a count of evidence markers, using a keyword-stem method adapted directly from Priniski and Horne's (2018) own code, which constitutes a direct instance of code reuse, not an independent re-implementation; the output of a HuggingFace persuasive-language detector (`chreh/persuasive_language_detector`); and document embeddings from the sentence-embedding model `BAAI/bge-base-en-v1.5` (native dimensionality 768), reduced via principal component analysis to 187 components, the smallest number retaining at least 85% of cumulative explained variance. All infrastructure for calling the LLM — client configuration, retry and timeout handling, checkpointed batch scoring, and prompt templates — is centralized in a single shared module, reused unchanged across the feature-engineering, rhetorical-scoring, and external-validation notebooks; this centralization is itself a modest but genuine contribution to the reproducibility of the pipeline.

The relational-feature notebook computes, for each numeric or categorical comment-side feature that has a natural original-post-side counterpart, both the original-post value and a derived "difference" (for numeric features) or "match" (a boolean agreement indicator, for categorical or near-binary features) quantity. The resulting representation therefore describes not only properties of the reply in isolation, but also how the reply's measured properties relate to those of the post it is replying to.

Combining these elements yields 41 handcrafted features: 22 numeric features spanning comment-side, original-post-side, and difference variants of sentiment, rhetorical-appeal, and readability measures together with evidence-marker counts; 6 boolean comment–post match indicators; 3 additional scalar features (two formality indicators and the persuasive-language detector output); and 10 one-hot-encoded columns produced by encoding the two six-category tone labels (comment-side and original-post-side), with the first category of each dropped as the implicit reference level — standard practice to avoid the linear dependency a full one-hot encoding would introduce, since a row's remaining encoded columns already determine membership in the dropped category when all of them are zero. Together with the 187 embedding components, this constitutes a 228-feature "trusted" representation.

A further six features — a lexicon-based sentiment score, the earlier (rejected-in-favor-of-LLM) categorical tone and style labels, and the earlier-generation ethos, pathos, and logos scores — were formally flagged as **untrusted**, reflecting the diagnosed shortcomings described above, and recorded as such in a dedicated file. The rationale for retaining instead of discarding them was to allow their marginal predictive value to be tested empirically during model selection (§2.3), instead of excluding them by assumption. Encoding these six raw features (two of which are categorical and together expand into more columns than one each) adds 11 columns, yielding a 239-feature "trusted+untrusted" representation — the 41 trusted and 11 untrusted handcrafted columns together total 52 handcrafted features, matching the "52 handcrafted features" figure used in the project's presentation, which does not itself separate the trusted and untrusted subsets. An "embeddings-only" representation (the 187 components alone, with no handcrafted features) was also constructed, to assess how much predictive value the handcrafted features add over embeddings alone. All three representations were carried forward as candidate feature sets into model selection.

Table 2 (Appendix A, split across Table 2a–2c) summarizes every feature family introduced in this section: its variants, source notebook(s), column count, and trusted/untrusted status.

2.3 Model Training and Evaluation Methodology

The prediction task is binary classification: given the representation of a comment (and, depending on feature-set variant, its relation to the original post), predict whether it was delta-awarded. Evaluation uses a thread-grouped 80/20 split (`GroupShuffleSplit`, grouped by discussion thread), assigning 3,262 comments from 320 threads to training and 709 comments from 81 threads to a held-out test set, with zero thread overlap between the two, confirmed programmatically. Grouping the split by thread matters because comments from the same thread share context — the same original post, often overlapping argumentative content, and sometimes near-duplicate phrasing across replies — so a standard random split risks leaking thread-specific information between training and test examples, inflating apparent performance in a way that would not generalize to genuinely unseen discussions.

This split protects against thread-level leakage specifically; it does not retroactively correct a separate limitation in how two of §2.2′s feature-engineering steps were fit. The categorical one-hot encoder (comment and original-post tone and style labels) and the PCA reduction of the document embeddings were both fit on the full, unsplit dataset during feature engineering, before any train/test division existed, here and in every notebook that rebuilds this feature matrix. Neither step uses the `is_convincing` label — both are unsupervised transformations of the input text alone — so this does not constitute label leakage, but it is a methodological limitation: the fitted encoding and the embeddings' principal-component basis were informed by feature values from rows that later became test-set rows, rather than from training data alone.

Model and feature-set selection used a further, inner split carved from the training data using the same thread-grouping logic (2,629 comments from 256 threads for inner training; 633 comments from 64 threads for validation), so that the 709-comment test set was not consulted during selection. A grid of 27 combinations was evaluated on this inner validation split: three classifiers (logistic regression, random forest, and XGBoost), three resampling conditions applied to the inner-training partition (no resampling, random undersampling, and SMOTE oversampling), and the three feature-set variants from §2.2 (trusted, trusted+untrusted, embeddings-only). SMOTE and random undersampling were included because the positive class is a minority (roughly 16% of the full dataset); resampling strategies are a standard response to this kind of imbalance, and including "no resampling" as a condition allows their marginal value to be assessed empirically.

The primary selection and evaluation metric throughout this project is PR-AUC (average precision), not accuracy or ROC-AUC. This choice follows directly from the class imbalance: with positive examples at roughly 16–20% depending on the specific split, a classifier that never predicts the positive class can already achieve high accuracy, and ROC-AUC can look favorable even when precision on the minority class is poor, because ROC-AUC is computed against the (large) majority class's true-negative rate. PR-AUC is sensitive specifically to the precision–recall trade-off on the minority class and is the more informative metric under this kind of imbalance. PR-AUC's no-skill reference value is the positive-class prevalence: for any classifier whose score does not discriminate between the two classes at all — the same score, or scores drawn without regard to the true label, assigned to every instance — the area under the

resulting precision–recall curve reduces exactly to that fold's own positive rate. In this project, that reference is computed for a given evaluation fold via a `DummyClassifier` configured to the `most_frequent` strategy (labeled "majority-class" in this report's tables): its `predict_proba` output is a single constant value for every instance in the fold, so its PR-AUC is numerically exactly equal to the fold's own positive rate. This is a consequence of the score being non-discriminating, not of the classifier's hard class decision; a classifier evaluated only on its hard 0/1 label — one that always outputs the negative label and never predicts the positive class at any threshold — has recall 0 and an undefined (or, by convention, zero) precision at its one operating point, not a precision equal to the positive rate. A second dummy baseline, `DummyClassifier(strategy="stratified")`, instead draws a random hard guess per instance matching the training class frequencies; on the 709-comment home test fold this reaches PR-AUC 0.2013, close to but not exactly equal to that fold's own 0.2031 positive rate, since it is an actual random classifier rather than a constant, non-discriminating score. Because the train/test split is grouped by thread, not stratified by label, a given fold's own positive rate can differ somewhat from the full dataset's 16.1% rate (the 709-comment test fold used throughout this report has a positive rate of approximately 20.3%), and the appropriate no-skill baseline for any PR-AUC reported later in this report is that specific fold's own rate, not the whole-dataset rate.

Selection on the inner validation split identified XGBoost with SMOTE oversampling, on the 239-feature trusted+untrusted representation, as the best-performing combination. This configuration was then evaluated once on the held-out 709-comment test set, achieving PR-AUC 0.3848. This number, and its subsequent revision after the length confound described in §2.4 was discovered, together motivate the remainder of this chapter.

Table 3. The full 27-combination inner-validation grid, sorted by PR-AUC. These are **validation**-set scores, used only for selection — not the 0.3848 **test**-set PR-AUC reported above for the selected configuration, which this grid does not itself contain (the selected row's own validation PR-AUC is 0.3063; the test-set figure comes from a separate, one-time evaluation after selection, described in the paragraph above). ROC-AUC is included as the secondary metric discussed earlier in this section; PR-AUC alone decided selection.

Classifier	Resampling	Feature set	Validation PR-AUC	Validation ROC-AUC
XGBoost	**SMOTE oversampling**	**trusted+untrusted**	**0.3063**	**0.6952**
Random Forest	SMOTE oversampling	embeddings-only	0.3043	0.6455
Random Forest	SMOTE oversampling	trusted+untrusted	0.2961	0.7088

Table 3 (continued).

Classifier	Resampling	Feature set	Validation PR-AUC	Validation ROC-AUC
Random Forest	Random undersampling	trusted+untrusted	0.2957	0.6933
XGBoost	SMOTE oversampling	embeddings-only	0.2952	0.6551
Random Forest	Random undersampling	trusted	0.2886	0.6989
Random Forest	SMOTE oversampling	trusted	0.2879	0.6999
XGBoost	No resampling	trusted	0.2845	0.6983
XGBoost	No resampling	trusted+untrusted	0.2816	0.7070
Random Forest	Random undersampling	embeddings-only	0.2673	0.6413
XGBoost	Random undersampling	trusted	0.2628	0.6659
Logistic Regression	SMOTE oversampling	trusted+untrusted	0.2574	0.6774
Logistic Regression	SMOTE oversampling	trusted	0.2530	0.6748
XGBoost	Random undersampling	embeddings-only	0.2485	0.6140
XGBoost	Random undersampling	trusted+untrusted	0.2461	0.6308
XGBoost	SMOTE oversampling	trusted	0.2436	0.6411

Table 3 (continued).

Classifier	Resampling	Feature set	Validation PR-AUC	Validation ROC-AUC
Random Forest	No resampling	trusted	0.2420	0.6157
Logistic Regression	No resampling	trusted	0.2353	0.6587
Logistic Regression	No resampling	trusted+untrusted	0.2347	0.6589
Random Forest	No resampling	trusted+untrusted	0.2285	0.6082
Logistic Regression	SMOTE oversampling	embeddings-only	0.2285	0.6473
XGBoost	No resampling	embeddings-only	0.2224	0.5963
Logistic Regression	No resampling	embeddings-only	0.2095	0.6302
Logistic Regression	Random undersampling	trusted	0.1816	0.5537
Logistic Regression	Random undersampling	trusted+untrusted	0.1814	0.5541
Logistic Regression	Random undersampling	embeddings-only	0.1786	0.5477
Random Forest	No resampling	embeddings-only	0.1506	0.4468

This validation fold's own no-skill PR-AUC baseline is 0.1690 (its own positive rate — not the whole-dataset rate of 0.1614, which this fold's rate differs from, consistent with how a thread-grouped split's fold-specific positive rate generally departs somewhat from the whole-dataset rate elsewhere in this report). All but one of the 27 combinations beat it; the sole exception is the weakest row, `RF (no resampling) | embeddings-only` (0.1506), which falls below both this fold's own baseline (0.1690) and the whole-dataset rate (0.1614) — the only combination in the grid that fails to beat no-skill under either baseline. `RF (oversampling) | embeddings-only` (0.3043) trails the selected configuration (0.3063) by 0.002 PR-AUC. For `XGB (oversampling)` specifically, PR-AUC on `trusted+untrusted` exceeds PR-AUC on `trusted` alone by 0.063 — the largest such difference among the 9 classifier/resampling pairs.

Column-order sensitivity. The `XGB (oversampling)` architecture used throughout this chapter is sensitive to the order its input columns are presented in, not just which columns are included. XGBoost (Chen & Guestrin, 2016) builds each decision tree by selecting, at every split, whichever candidate feature yields the greatest reduction in training loss; when two or more features are correlated — common throughout this project's feature sets, from the 187 embedding components to the engineered features' own overlapping signals — their split-gain values can come out essentially tied, and such ties are broken deterministically

by each feature's position (column order) in the input matrix, not by anything about the feature's actual values. This has been found to swing PR-AUC by as much as 0.0482 on one of this project's larger feature representations (the 237-column set tested in §2.7), purely from reordering the same columns. The sensitivity is specific to the tree-building step, not `SMOTE`'s (Chawla et al., 2002) oversampling: `SMOTE`'s neighbor-finding operates on Euclidean distance, which does not depend on the order features are summed in. Because of this, every PR-AUC value reported from this point in the chapter onward **for an evaluation of this order-sensitive `XGB (oversampling)` architecture** is computed as a mean and standard deviation across eight column orderings (the original ordering plus seven random reshufflings) of the same underlying feature matrix, not as a single model fit. This does not apply to the logistic-regression-based baselines (TF-IDF, `word_count` alone) or the dummy classifiers compared against it throughout this report: those models have no column-order degree of freedom — a convex solver and a label-only classifier are both order-invariant by construction — and continue to be reported as single-fit point estimates, typically alongside a bootstrap 95% confidence interval instead (§2.4). These standard deviations describe sensitivity to an essentially arbitrary implementation detail — column order — and are not bootstrap confidence intervals or any other estimate of sampling uncertainty; they should not be interpreted as such.

2.4 Baseline Investigation and the Length Confound

Having selected and evaluated the engineered-feature pipeline, a substantially simpler baseline was evaluated on the identical 709-comment test fold, as a check on whether the engineered representation was adding value over a much cheaper alternative: TF-IDF vectorization of the comment text alone, with a logistic regression classifier and no engineered features at all. This baseline outperformed the selected 239-feature engineered pipeline on the same test fold, reaching PR-AUC 0.5119 against the engineered pipeline's 0.3848.

This result prompted a diagnostic step: examining which tokens the TF-IDF model weighted most heavily revealed that the great majority were common stopwords, together with the token "edit" (characteristic of Reddit's edit-marker convention) — a pattern more consistent with the model tracking how much text a comment contains than with it tracking rhetorical content.

This diagnostic finding motivated testing the role of comment length directly, in two ways. First, a second, even simpler baseline was evaluated: a single-feature logistic regression using only the comment's raw word count. This baseline outperformed TF-IDF too, reaching PR-AUC 0.5256 — the strongest of the three models compared so far, using the least information.

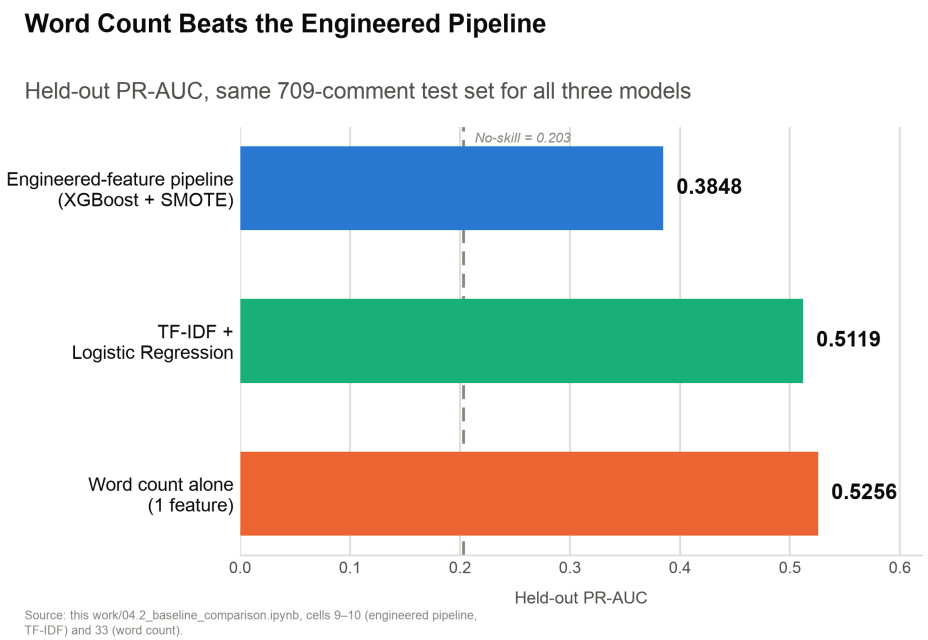


Figure 2. Held-out PR-AUC for the originally selected engineered-feature pipeline, a TF-IDF baseline, and a word-count-only baseline, evaluated on the identical 709-comment test fold (`04.2_baseline_comparison.ipynb`).

Second, the raw relationship between comment length and outcome was characterized directly: comment length and outcome are strongly associated in this dataset, with a mean word count of 78.0 for non-convincing comments and 216.7 for convincing comments, a point-biserial correlation of 0.3716 (denoted r hereafter). To rule out the possibility that this association was purely an artifact of some discussion threads being systematically longer or more active than others (and hence containing both longer comments and more opportunities for a delta), a within-thread check was performed: restricting to the 200 threads that contained at least one comment of each outcome class, the convincing comment was longer than that same thread's own average non-convincing comment in 153 of the 200 threads (76.5%).

In response to this finding, `word_count` was added to the engineered feature set as an additional column, and the full 27-combination selection grid from §2.3 was rerun from scratch on the word-count-augmented feature sets, instead of simply appending the feature to the previously selected configuration. The same architecture was selected again — XGBoost with SMOTE oversampling on the trusted+untrusted feature set, now including `word_count` — and this recovered most of the shortfall relative to the simpler baselines: test PR-AUC rose from 0.3848 to 0.4922.

These two figures are each the mean of multiple executions across eight column orderings, not a single model fit, per the column-order-sensitivity convention established in §2.3. This gives 0.3676 ± 0.0182 for the originally selected pipeline and 0.4831 ± 0.0257 after the word-count-augmented rerun, with bootstrap 95% confidence intervals of [0.311, 0.463] and [0.410, 0.586] respectively. The remainder of this report treats the recognition and correction of this length confound as a methodological turning point: it reframes the project's central question from whether textual features predict delta awards to which predictive signal, if any, remains once comment length is accounted for — the question addressed in §2.5 onward.

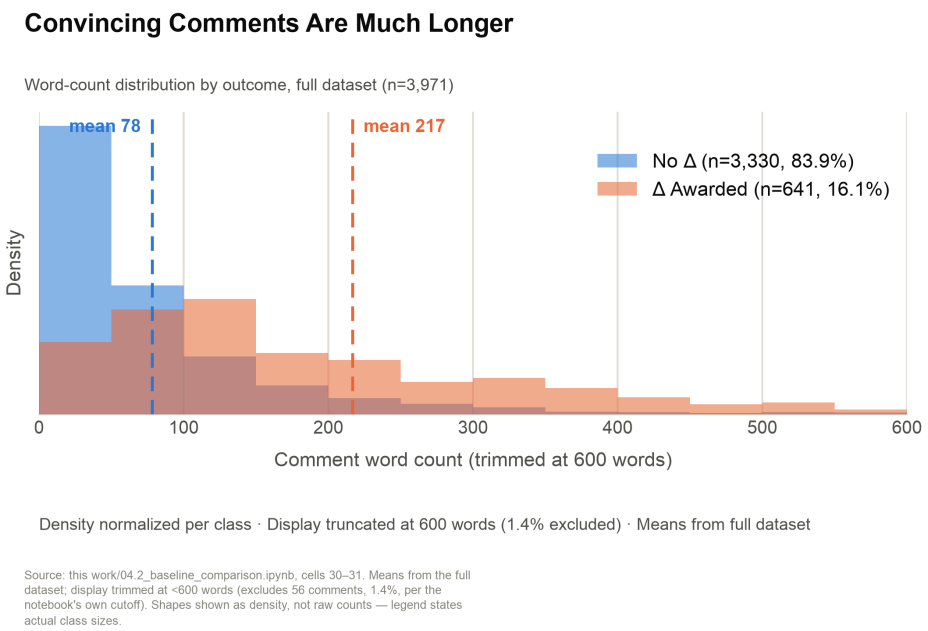


Figure 3. Word-count distribution by outcome for the full 3,971-comment dataset, with each class's mean indicated (`04.2_baseline_comparison.ipynb`).

These four models — both their single-fit point estimates and, for the order-sensitive engineered pipeline, its order-robust mean ± SD — are tabulated together in the Results chapter (Table 8, §3.1.2; the word-count-augmented re-selected pipeline's figures are discussed in §3.1.4), rather than repeated here.

2.5 Length-Controlled Analysis

Folding `word_count` into the feature set and reselecting recovers predictive performance, but does not on its own establish whether any part of the recovered performance reflects something beyond length — in principle, a model's entire predictive value could still reduce to "longer comments are more often convincing." A dedicated length-controlled analysis was carried out on the full 3,971-comment dataset to address this directly, using two different constructions to reduce the length imbalance between outcome classes: a loose word-count floor (84 words — the convincing group's own 25th percentile), which discards the shortest non-convincing comments and yields a sample of 1,638 comments, and a strict decile-bin length match, which bins convincing comments' `word_count` into ten equal-count groups and samples a matching number of non-convincing comments from each group, yielding a sample of 1,235 comments. Both samples are constructed using the very outcome label and length values later evaluated, before being split, by the same thread-grouped methodology as §2.3, into train and test partitions; each resulting test partition is accordingly a partition of a controlled, outcome-balanced evaluation sample, not a naturally distributed held-out population, and its own no-skill baseline differs from the full dataset's for that reason. For each of these two samples, and for the full, unmatched dataset, the lift of several models — TF-IDF, word-count-only, and the full engineered pipeline — over that sample's own no-skill baseline was computed, allowing each model's apparent skill before and after length control to be compared directly.

Across all three models tested, apparent predictive lift drops substantially once length is controlled for, but in no case falls to zero or below — some predictive signal beyond raw length survives length control for every approach tested, though modest in absolute terms; the exact lift figures for each sample are reported in Results §3.1.4. A further, separately noted methodological result from this analysis concerns the risk of re-selecting a model on a small sample: re-running the complete 27-combination selection grid from §2.3 from scratch on the small (1,235-row) strictly length-matched sample selected a different architecture that, when evaluated on the corresponding test partition, scored *below* its own no-skill baseline — a concrete instance of overfitting attributable to comparing many candidate configurations (27) against a small validation partition (205 rows). By contrast, simply reusing the architecture already selected on the full dataset, without re-selecting, scored positively on the same matched test partition. This result is presented as a specific, worked illustration of a general methodological risk — model re-selection on small samples — and not as evidence against re-selection as a practice in general.

Table 4. The two length-controlled samples used in this section, side by side (both built in `04.3_length_controlled_analysis.ipynb`, from the full 3,971-comment dataset).

Sample	Matching procedure	n	`word_count` r (unmatched → matched)
Loose length floor	Keep every convincing comment; restrict non-convincing comments to ≥ the convincing group's 25th-percentile length (84 words)	1,638	0.372 → 0.093
Strict decile-bin match	Decile-bin match: convincing comments' `word_count` split into 10 equal-count bins; non-convincing comments sampled from the matching bin	1,235	0.372 → 0.066

2.6 Rhetorical-Strategy Scoring

The length-controlled analysis in §2.5 establishes that some non-length signal survives across several modeling approaches, but it does not characterize what that signal consists of. To investigate whether an argument's manner of construction carries predictive information beyond simply how much is written, eight rhetorical strategies relevant to CMV-style argumentation were defined: justification, concession, evidence, refutation, personal narrative, call-to-action, simplification, and question-asking. Each strategy is scored on an ordinal 1–5 scale, independently of the others, by an LLM (Gemini). The general approach — using an LLM to assign numeric persuasion-strategy scores to naturally occurring text — follows the methodology described by Labruna, Modzelewski, Satta, and Da San Martino (2026), who applied an analogous scoring approach primarily to the Winning Arguments, Anthropic/Persuasion, and Persuasion for Good corpora. The eight strategies used in this project were defined independently for the CMV setting and are not a reuse of Labruna et al.'s own six-category taxonomy, although two strategy names (justification, simplification) happen to be shared.

Scoring was restricted to comments of at least 30 words, or of any length if the comment was delta-awarded, yielding a scored population of 2,753 comments across 400 original posts — a subset of the full 3,971-comment dataset, not the full dataset itself. This restriction reflects a judgment that very short comments carry too little content for a meaningful rhetorical-strategy judgment, while the "or delta-awarded" clause ensures that no positive example is discarded regardless of its length. One consequence of that clause is worth stating directly: 42 of the 641 delta-awarded comments (6.6%) are themselves under 30 words, the shortest at 2 words, and were scored on the full eight-strategy rubric regardless — a rubric built for comments with enough content to exhibit more than one strategy at once has correspondingly little to work with on these particular rows. Because the restriction disproportionately removes short, non-convincing comments while retaining every convincing comment, the resulting subset's convincing-comment prevalence (641 of 2,753 comments, 23.3%) is substantially higher than the full dataset's 16.1%. Later sections that report figures computed within this scored subset (§2.7 onward) use this subset's own denominator, not the full dataset's 16.1% — the two should not be read as directly comparable.

The scoring prompt went through one documented revision. An initial version presented each comment to the LLM without including the original post it was replying to. Review of this initial pass identified that two of the eight strategies — concession and refutation — are defined relative to a specific point made in the original post, and cannot be judged accurately without the LLM having access to that point. A revised prompt batches comments by original post and includes the post's own text as context. In a manually reviewed comparison sample, refutation scores increased for the majority of comments under the revised prompt (6 of 14 reviewed rows increased, 1 decreased), consistent with the earlier, original-post-blind prompt having systematically under-detected refutation. The full scoring run used the revised, original-post-aware prompt.

Two reliability checks were carried out, both explicitly framed as lightweight checks, not a formal inter-annotator-agreement study: a test-retest check (200 comments scored twice, independently, and the two passes compared) and a comparison against a single human judge's hand-scoring of 18 comments (9 convincing, 9 not). The human-agreement figures are the judge's final scores, after she reviewed her own first-pass disagreements with the LLM and corrected some of them; initial agreement was considerably weaker (exact-match as low as 11.1%, for justification, before this review). Agreement is strong on the test-retest check across all eight strategies, and weaker but still reasonable on the human-agreement check, worst specifically for refutation and justification — the two hardest strategies to calibrate against human judgment even after review. This reliability pattern is directly relevant to results reported later in this project, since refutation and justification also prove to be the two strategies most central to the project's predictive and interpretive findings (§2.8). Exact reliability figures for each strategy are given in Results, Table 10 (§3.2.1).

Table 5. The eight rhetorical strategies and their definitions.

Strategy	Definition
Justification	Gives a reason or logical basis for the position, beyond simply asserting it
Concession	Acknowledges a valid point, limitation, or partial truth in the opposing view before responding
Evidence	Cites a concrete fact, statistic, study, source, or specific example to support a claim
Refutation	Directly identifies and counters a specific flaw, assumption, or weak point in the original argument
Personal narrative	Draws on the commenter's own lived experience to make the case
Call-to-action	Offers a concrete alternative way to act on or think about the issue, not just criticism
Simplification	Uses an analogy, thought experiment, or simplified restatement to make an abstract point concrete
Question-asking	Poses a genuine question back to the original poster that probes an assumption or invites clarification

2.7 Strategy Ablation

To test whether the eight rhetorical-strategy scores add predictive information beyond an existing engineered-feature representation, the scores were appended to an engineered-feature baseline and the combined representation was evaluated using the same thread-grouped evaluation logic used elsewhere in this project, restricted to the 2,753-comment rhetorical-scoring population from §2.6. The engineered-feature baseline used in this analysis is the 228-feature trusted representation from §2.2 with `word_count` added (229 columns in total); it does not include the eleven untrusted-derived columns that formed part of the 239- and 240-feature representations selected and evaluated in §2.3 and §2.4. This is noted explicitly because it means the ablation described in this section is not a direct continuation of the exact feature set carried through §2.3–§2.4, but a related, narrower representation built specifically for this analysis. Combined with the eight strategy scores, the resulting representation totals 237 columns.

Adding the eight strategy scores to the engineered-feature baseline produces a real, order-robust gain over that baseline alone; the strategy scores alone, with no engineered features at all, also carry a real, standalone predictive signal, though a smaller one than the combined representation. Exact figures are reported in Results, Table 11 (§3.2.2).

A further, parsimony-oriented finding is reported here because it bears on how the engineered and strategy-score features interact. Among all feature combinations evaluated, the smallest — the eight strategy scores together with `word_count` alone (nine columns in total) — achieved the highest PR-AUC of any configuration tested in this project, 0.5379 (± 0.0109), exceeding the full 237-column representation. Re-running the classifier and resampling grid restricted to these nine columns found that a random forest classifier performed comparably

to the originally selected XGBoost configuration (0.5432 ± 0.0065 versus 0.5380 ± 0.0108) — a difference smaller than either classifier's own order-induced spread, and therefore not a basis for preferring one architecture over the other on this particular feature set.

Unlike the classifier-and-resampling grid search in §2.3, which uses a dedicated inner-validation split the test set never touches, this feature-family comparison was evaluated directly against this section's own held-out test partition, across every candidate feature-set configuration considered. The nine-column combination is accordingly the product of an exploratory, test-informed search over feature combinations rather than a single evaluation of a feature set chosen independently of the test data; its reported PR-AUC should be read with that qualification in mind.

This gain's survival under length control is the most direct test of this project's central claim — that rhetorical-strategy scores carry real signal beyond comment length, not just another proxy for it — mirroring the same test §2.5 applied to `word_count`'s own contribution. It was checked using a length-matched sample built specifically for this 2,753-comment population: the same decile-bin matching procedure used for `04.3`'s strict decile-bin match (§2.5) was independently re-run here, yielding 1,235 comments. On this length-matched sample, the strategy-score gain retains a substantially larger share of its full-population value than `word_count`'s own lift or the engineered pipeline's own lift retained under `04.3`'s strict matching (§2.5); exact figures are reported in Results, Table 12 (§3.2.3). This indicates the strategy-score signal is not primarily a proxy for comment length, unlike previous signal examined.

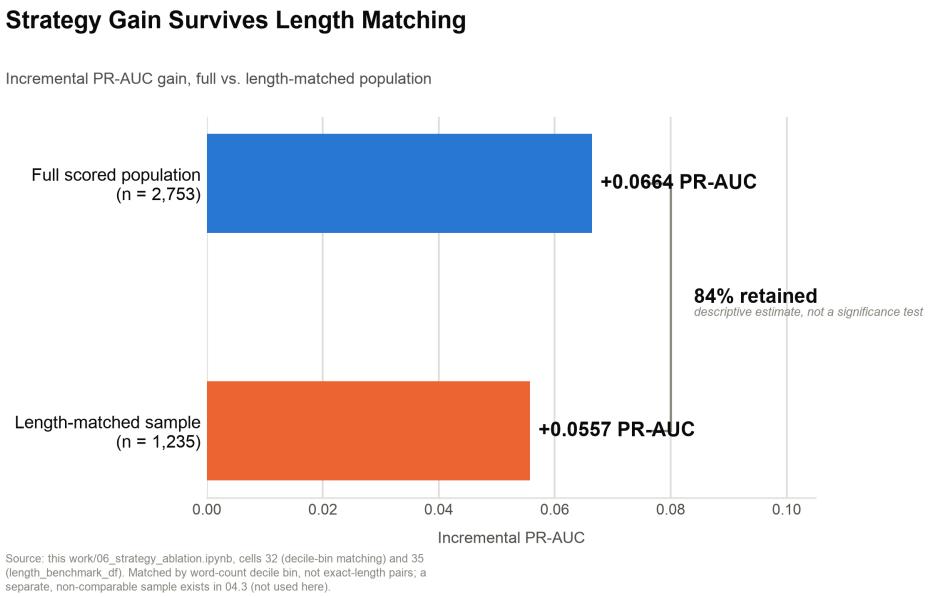


Figure 4. The strategy-score PR-AUC gain reported above, recomputed on the length-matched sample within the 2,753-comment scored population, showing how much of the gain survives length control (`06_strategy_ablation.ipynb`).

2.8 Strategy Interpretation

To identify which individual strategies underlie the aggregate gain reported in §2.7, two complementary interpretive analyses were applied to the nine-column model (the eight strategy scores plus `word_count`) identified in that section as the strongest-performing combination.

The first is a SHAP (SHapley Additive exPlanations; Lundberg & Lee, 2017) analysis, using a tree explainer applied directly to the fitted XGBoost model, which summarizes how much each feature moves the model's output, on average, across the comments in a given evaluation fold. The second is a model-free univariate comparison: for each strategy, a two-sided Mann–Whitney U test compares the distribution of that strategy's scores between convincing and non-convincing comments, converted to a rank-biserial correlation as the corresponding effect size; p-values across the eight simultaneous tests are corrected for multiple comparisons using the Benjamini–Hochberg false-discovery-rate procedure, with significance assessed at a corrected threshold of 0.05. Both analyses were computed twice: once on the full 2,753-comment scored population, and once on the strict, decile-bin length-matched sample described in §2.7, to distinguish which strategies' apparent importance holds up once length is controlled for from those whose apparent importance is better explained by their correlation with length.

SHAP values and the Mann–Whitney rank-biserial effect size answer related but distinct questions: the former describes how much a specific fitted model relies on a feature, while the latter describes a model-independent, univariate association between a feature's score distribution and the outcome. Neither is a causal-effect estimate — both describe observed association or model reliance, not a demonstrated causal mechanism — and the two measures are not guaranteed to agree with one another, nor are they intended as interchangeable. In this project, these two measures produce a genuine point of disagreement about which single strategy is most important even on the home dataset's full population; this is reported directly, and the Discussion chapter explains why the two measures disagree — reflecting their different analytical roles (model-dependent predictive attribution versus model-free marginal association) rather than an error in either computation — instead of resolving the disagreement by relying on only one of the two measures.

Both interpretive measures, for all eight strategies, full population and length-matched sample, are reported together in Results, Table 13 (§3.2.5).

One row is worth flagging, not leaving implicit: `personal_narrative`'s rank-biserial effect is not significant on the full population but *is* significant, and negative, on the length-matched sample ($r = -0.0859$) — the only strategy whose significance status and sign both shift between the two populations.

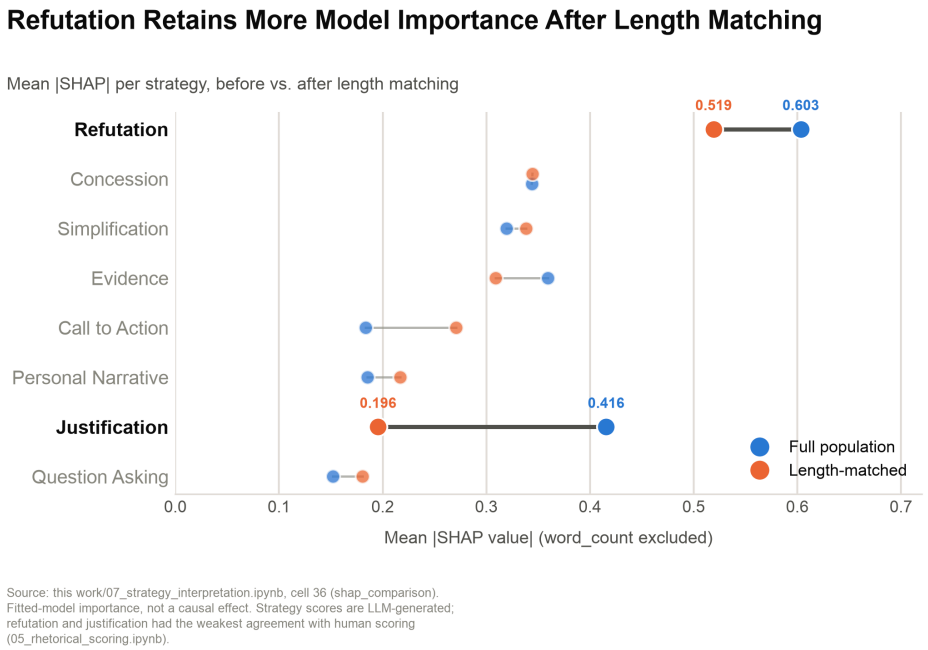


Figure 5. Mean |SHAP| value for each of the eight rhetorical strategies, full population versus length-matched sample, for the nine-column selected model (`07_strategy_interpretation.ipynb`).

The two measures above converge on the same three strategies — refutation, justification, and simplification — as the ones whose rank-biserial effect size remains statistically significant both before and after length matching. A pruning check tested this directly: dropping the remaining five strategies from the nine-column model, individually, in combination, and all at once, against the same test partition used throughout this section. The four-column combination that survives this pruning — these three strategies plus `word_count` — is referred to as the robust core in the remainder of this report. Its selection differs methodologically from the nine-column search in §2.7: the robust core's three strategies were fixed in advance by the Mann-Whitney significance criterion above, not chosen by searching for the best-scoring subset — though that criterion was itself computed on the same test partition instead of an independent validation sample, so its selection is not fully insulated from the test set either, just in this narrower, criterion-based sense rather than via a PR-AUC search. Exact PR-AUC figures for both compact configurations are reported in Results, §3.2.4.

2.9 External Validation on Winning Arguments

To assess whether the rhetorical-strategy findings reported in §2.7–§2.8 are specific to this project's own dataset-construction choices, the same scoring rubric and modeling approach were applied to Winning Arguments (WA; Tan, Niculae, Danescu-Niculescu-Mizil, and Lee, 2016), an independently constructed persuasion corpus built from paired winning and losing replies on r/changemyview, accessed through ConvoKit. WA differs from the home dataset in an important respect: its losing (non-winning) replies are deliberately selected to be topically similar and substantively comparable to the corresponding winning reply, matched by Jaccard similarity within the same thread, whereas the home dataset's non-convincing comments are simply the other, unfiltered replies already present in the same threads (§2.1). This difference in how negative examples are constructed is a methodologically important point of contrast

between the two datasets, revisited in the Results and Discussion chapters. The two datasets also differ in which replies are eligible at all: WA's winner/loser pairs are restricted to direct top-level replies to the original post (exactly one level below it), whereas the home dataset includes every reply in a thread regardless of its nesting depth.

A total of 8,106 comments, spanning 3,051 original posts, were collected and scored on the same eight-strategy rubric, using the same scoring infrastructure described in §2.6. Ten comments were subsequently dropped following an unparseable response from the LLM, leaving 8,096 scored comments. These were evaluated on WA's own native training and held-out partitions — the same partition the corpus is distributed with, and the same partition on which its own published benchmark figures are computed — comprising 6,584 comments for training and 1,512 for evaluation.

The model architecture and training configuration already selected on the home dataset (`RobustScaler`, SMOTE oversampling, and XGBoost) were reused unchanged on WA, with no new architecture selection, hyperparameter search, or feature-set tuning performed using WA data. A new model instance was then fit on WA's own native training split and evaluated on WA's own native held-out split. This is a transfer of the modeling procedure — reusing a previously selected configuration but fitting new parameters on the new dataset's own training data. Because WA was scored only on the eight-strategy rubric described above, not the full engineered-feature pipeline from §2.2, the feature set used for this fit is the nine-column model (the eight strategy scores plus `word_count`) identified in §2.7 as the home dataset's own strongest-performing combination, not the larger 237-column representation. A smaller set of narrower feature-set configurations (`word_count` alone, the eight strategy scores alone, and other subsets of the nine columns) was also evaluated on WA using the same order-robust procedure as §2.7, to check whether the representation selected on the home dataset is also the strongest-performing one on an independently constructed dataset.

The same two interpretive analyses described in §2.8 — a SHAP tree-explainer analysis and a Mann–Whitney rank-biserial effect-size comparison, each computed on the full scored population and on a length-matched sample — were repeated on WA, fitted to this same nine-column model, to compare which individual strategies the two analyses rank as most important on WA against the ranking already reported for the home dataset in §2.8. The resulting cross-dataset strategy-importance comparison is reported in the Results chapter, not here.

Because that comparison surfaces a reversal in which strategy ranks highest (§2.8), an obvious question is whether a strategy's score is simply less reliably assigned on WA than on the home dataset. The same test-retest design used for the home dataset's own reliability check (§2.6) was therefore repeated here: 200 WA comments were scored twice, independently, and the two passes compared. Every strategy proved more internally consistent on WA than on the home dataset, finding no straightforward evidence that scoring instability explains the ranking differences reported in the Results chapter — though this check does not rule out other, more systematic measurement differences, such as the LLM interpreting the same rubric somewhat differently against WA's text distribution than against the home dataset's.

A length-matched robustness check was also performed on WA, using the same decile-bin matching procedure as §2.5/§2.7, adapted to WA's own correlation between `word_count` and outcome — considerably weaker than the home dataset's own confound, since WA's winning

and losing replies were already fairly similar in length before any matching, by construction. This check compares `word_count` alone against `word_count` combined with the eight strategy scores, not the home dataset's own comparison of a full engineered-feature pipeline with and without those scores, since WA was never scored on that pipeline. Exact figures are reported in Results, §3.3.2.

Because WA's class balance and negative-sampling procedure differ substantially from the home dataset's, raw PR-AUC values are not directly comparable between the two datasets without reference to each dataset's own no-skill baseline; WA's held-out fold has a no-skill PR-AUC of 0.5291, reflecting its near-balanced, pairwise construction, compared with the considerably lower no-skill baselines (roughly 0.20–0.24 depending on the specific fold) reported for the home dataset elsewhere in this chapter. Pairwise accuracy — the proportion of winner/loser pairs for which the model scores the actual winning reply higher than the actual losing reply — is also reported for WA, since it is the metric reported in the published literature this analysis is compared against (Labruna et al., 2026). WA's own pairing procedure yields 4,263 winner/loser pairs overall, of which 807 fall in the held-out split — the same heldout pair count reported in the published literature. Of the ten comments dropped earlier for an unparseable LLM response, four fall in the held-out split, and each belongs to a different pair whose other side was also dropped, removing two complete pairs from the held-out split; the remaining 805 held-out pairs, both sides scored, are used to compute this metric.

One limitation of this external validation is stated directly here, because it bears on how the WA results should be read, not just a matter for the Discussion chapter: this analysis supports the broader claim that rhetorical-strategy scores carry predictive information in a second, independently constructed dataset, and the length-matched check above finds a broadly similar pattern of retention to the home dataset's own ablation. However, because that check is narrower in scope — comparing `word_count` alone against `word_count` plus strategy scores, not the home dataset's own full engineered-feature pipeline with and without those scores — it does not, by itself, establish that the *incremental* predictive contribution of rhetorical strategies specifically beyond comment length replicates to the same extent documented for the home dataset in §2.7. Establishing that stronger, more specific claim would require a directly comparable ablation, which falls outside the scope of the present external-validation check.

Table 6. How each dataset's underlying corpus was constructed, side by side — the negative-example row is the single most consequential difference between them, discussed above.

	Home dataset	Winning Arguments (WA)
Source	r/changemyview, collected directly via the Reddit API (PRAW)	r/changemyview-derived, accessed via ConvoKit (Tan, Niculae, Danescu-Niculescu-Mizil, and Lee, 2016)
Unit of construction	Comment-level — not every reply becomes a row (see the two rows below)	Winner/loser pair-level — each winning reply is matched with one topically similar losing reply; an original post can contribute more than one pair

Table 6 (continued).

	Home dataset	Winning Arguments (WA)
Reply depth eligible	Any reply in the thread, regardless of nesting depth	Direct top-level replies to the original post only (exactly one level below it) — matching the original WA paper's own framing of a pair as two arguments made directly in response to the same post
Positive (convincing) examples	Comment awarded a delta, regardless of its position in the thread; its own descendant replies are discarded	Comment awarded a delta (therefore marked as the pair's winning reply)
Negative (non-convincing) examples	Only thread-ending *leaf* comments (no replies of their own) among the non-convincing comments; a non-convincing comment with further replies underneath it is excluded entirely, not included as its own row	The single reply most topically similar (Jaccard similarity) to the winner, among replies that did not win
Class balance	641 convincing (16.1%)	4,240 winning comments (52.3%), 3,866 losing comments — close to, but not exactly, balanced. Pairs themselves are exactly 1:1 by construction, but a losing reply can be the closest topical match for more than one winning pair and is stored once rather than once per pair (338 of 4,263 losing slots are such reused comments, versus only 22 of 4,263 winning slots), so deduplicating pairs down to individual comments leaves more unique winning comments than unique losing ones
Paired design	No — comments are evaluated individually, not as pairs	Yes — 4,263 winner/loser pairs total

Table 7. How this project used each dataset for the external-validation check described in this section — these are this project's own analysis choices, not properties of either corpus as originally constructed.

	Home dataset	Winning Arguments (WA)
Comments/ posts scored	2,753 comments, 400 original posts (§2.6 scored subset, not the full 3,971-comment dataset)	8,096 scored comments (8,106 collected, 10 dropped for unparseable LLM responses), 3,051 original posts
Train/test split	This project's own thread-grouped 80/20 split of the 2,753-comment scored population (§2.7): 2,161 train / 592 test — not the full-dataset 3,262/709 split used in §2.3–§2.4	WA's native distributed split (not re-split by this project): 6,584 train / 1,512 heldout
Positive rate actually scored/ evaluated	23.3% (641 of 2,753 — the §2.6 scored subset's own rate, not the full dataset's 16.1%)	52.9% in the heldout split used here, close to but not identical to the 52.3% rate across the whole WA corpus (Table 6) — this fold-to-fold variation (train alone is 52.2%) reflects how the reused losing comments happen to fall across original posts, not an effect of this project's train/heldout split or the ten dropped comments
No-skill PR-AUC baseline (this project's model)	0.2382 (the 592-comment test fold's own positive rate, §2.7)	0.5291 (heldout fold's own positive rate)
Pairs evaluated for pairwise accuracy	N/A — not a paired evaluation	805 of the native 807 held-out pairs — 2 pairs lost both sides to the unparseable-response drop

3. Results

This chapter reports the project's quantitative findings, organized around the three research questions introduced in Chapter 1: predictability and the role of comment length (RQ1), the incremental value and importance of rhetorical-strategy scores (RQ2), and cross-corpus generalization on an independently constructed dataset (RQ3). Procedural detail — how each dataset, sample, or model was constructed — is not repeated here except where needed to interpret a number; it is described in full in Chapter 2. Broader interpretation of these findings, their relationship to prior work, and their limitations are reserved for the Discussion chapter.

3.1 RQ1: Predictability and Comment Length

3.1.1 Performance of the engineered-feature pipeline

Model selection on the inner validation split (§2.3) identified XGBoost with SMOTE oversampling on the 239-feature trusted+untrusted representation as the best-performing combination, with validation PR-AUC 0.3063 against that fold's own no-skill baseline of 0.1690. Evaluated once on the 709-comment held-out test set, this configuration reached **PR-AUC 0.3848** (ROC-AUC 0.6982, F1 0.4046 at a validation-tuned threshold), against a test-fold no-skill baseline of 0.2031. This is a real improvement over no-skill, but it is the starting point the remainder of this section re-examines, not the project's final answer to RQ1.

3.1.2 Comparison with TF-IDF and word-count baselines

Two substantially simpler baselines, evaluated on the identical 709-comment test fold, both outperformed the selected engineered pipeline. A TF-IDF-vectorized bag-of-words representation with logistic regression, using no engineered features at all, reached PR-AUC 0.5119. A single-feature logistic regression using only the comment's raw word count reached PR-AUC 0.5256 — the strongest of the three models, using the least information.

Table 8. Held-out PR-AUC and ROC-AUC for the engineered-feature pipeline and the two baselines that motivated this section, on the identical 709-comment test fold (no-skill PR-AUC 0.2031). The engineered pipeline is the only order-sensitive row (§2.3, §2.7); its order-robust mean ± SD across eight column-order shuffles is given alongside its single-fit point estimate. The other three rows have no column-order degree of freedom and are reported as single-fit point estimates only.

Model	PR-AUC	PR-AUC (order-robust mean ± SD)	ROC-AUC	ROC-AUC (order-robust mean ± SD)
Majority-class (no-skill reference)	0.2031	order invariant	0.5000	order invariant
Selected engineered pipeline (XGBoost + SMOTE, trusted+untrusted)	0.3848	0.3676 ± 0.0182	0.6982	0.6903 ± 0.0096
TF-IDF + Logistic Regression	0.5119	order invariant	0.8150	order invariant
Word count alone (1 feature)	0.5256	order invariant	0.8294	order invariant

The "majority-class" row is this fold's own positive-class prevalence, not a property of a hard always-negative decision rule (§2.3). The three non-reference values are plotted in Figure 2 (§2.4).

Inspecting the TF-IDF model's coefficients showed that the great majority of its most heavily weighted tokens were common stopwords, together with the token "edit" — a pattern more consistent with the model tracking how much text a comment contains than with it tracking rhetorical content. A follow-up check, refitting TF-IDF with English stopwords removed, found that this did not close the gap to `word_count`: PR-AUC fell from 0.5119 to 0.4394, and the correlation between the model's predicted probability and raw word count only dropped from r = 0.8433 to r = 0.7417 — still far above the raw label's own correlation with length (r = 0.3716, §3.1.3) — so stopword removal did not cleanly separate a length-independent rhetorical signal from a length-proxy effect: the lost predictive power is at least as consistent with stopwords being an unusually reliable length-tracking channel — common function words appear in nearly every sentence and scale smoothly with length, unlike sparser content words — as it is with them holding genuinely separate, non-length content.

Two further qualifications apply to this comparison. First, the engineered pipeline's single-fit point estimate (0.3848) sits toward the high end of what the architecture actually produces across column orderings — the order-robust mean in Table 8 (0.3676 ± 0.0182) is about one SD lower — since XGBoost's tie-breaking among this feature set's many correlated columns is itself sensitive to column order, with the mechanism set out fully in §2.3. Second, a bootstrap-based reliability check (§2.3) gives 95% confidence intervals of [0.311, 0.463] for the engineered pipeline and [0.430, 0.603] for TF-IDF, which overlap substantially. The ranking "word count

beats TF-IDF beats the engineered pipeline" is this test fold's best point-estimate ordering, consistent across every variant of the comparison tried, but it is not a gap any single 709-comment split proves beyond its own sampling noise.

3.1.3 Comment-length distributions and within-thread evidence

Comment length and outcome are strongly associated in the full 3,971-comment dataset: convincing comments average 216.7 words against 78.0 for non-convincing comments, a point-biserial correlation of $r = 0.3716$ ($p = 2.92 \times 10^{-130}$). This distribution, by outcome, is shown in Figure 3 (§2.4).

Two within-thread checks rule out this association being purely an artifact of some threads being systematically longer or more active than others. A fixed-effects check, demeaning both word count and outcome by thread before correlating them, found the relationship weakens but survives: r drops from 0.3716 across the whole dataset to $r = 0.2909$ within threads ($p = 2.78 \times 10^{-78}$) — some of the raw correlation genuinely is thread-level, but a strong, highly significant relationship remains after removing it. A paired comparison restricted to the 200 threads (of 401) containing at least one comment of each outcome found the convincing comment longer than that same thread's own non-convincing average in 153 of the 200 threads (76.5%), by a median of 87.4 words (Wilcoxon signed-rank $p = 4.15 \times 10^{-19}$). Even comparing only within the same thread, on the same topic, the comment that earns the delta is usually the longer one.

3.1.4 Length-controlled predictive comparisons

Two length-matched samples were constructed from the full 3,971-comment dataset to test whether each model's apparent skill depends on length: a loose floor keeping every convincing comment and restricting non-convincing comments to at least the convincing group's own 25th-percentile length (84 words; 1,638 comments), and a strict decile-bin match sampling non-convincing comments from the same length deciles as convincing comments in matching counts (1,235 comments). Because both samples are themselves built from the outcome label and length values under study, their test partitions are controlled, outcome-balanced evaluation samples, not naturally distributed held-out populations (§2.5). For each sample, and for the full dataset, each model's PR-AUC lift over that sample's own test-fold no-skill baseline — not the whole-sample rate, which can drift from the actual test-fold rate on a split this small — was computed for TF-IDF, word count alone, and the engineered pipeline.

Table 9. PR-AUC lift over each sample's own test-fold no-skill baseline, full dataset versus the two length-controlled samples (`04.3_length_controlled_analysis.ipynb`).

Sample	n	Test-fold no-skill	TF-IDF lift	`word_count` lift	Engineered-pipeline lift
Full dataset	3,971	0.2031	0.3088	0.3225	0.1817
Loose length floor	1,638	0.3903	0.0478	0.0898	0.0165
Strict decile-bin match	1,235	0.4679	0.0767	0.0576	0.0334

Every model's lift collapses sharply once length stops separating the classes — by roughly three-quarters to seven-eighths of its full-dataset value, depending on the model and the

strictness of the match — but in no case falls to zero or below. `word_count`'s own lift falls to about 18% of its full-dataset value under the strict match (0.0576 / 0.3225), and the engineered pipeline's lift falls to a comparable 18% (0.0334 / 0.1817). An order-robust check on the engineered pipeline's own lift (mean over eight column orderings) moves these point estimates by at most roughly one order-noise standard deviation, in either direction depending on the sample, and does not change this qualitative pattern. Some predictive signal beyond raw length survives length control for every approach tested, though its degree is modest in absolute terms and the samples involved are small (1,235–1,638 rows).

Re-running the full 27-combination selection grid (§2.3) from scratch on the strict decile-matched sample illustrates a distinct methodological risk, not a substantive finding about length. The grid's winner by validation PR-AUC on this sample (`RF (no resampling) | embeddings-only`, validation PR-AUC 0.6656 against a 205-row validation set) scored PR-AUC 0.4518 on the corresponding test partition — *below* that partition's own no-skill baseline of 0.4679, a lift of −0.0162. Simply reusing the architecture already selected on the full dataset (XGBoost + SMOTE, trusted+untrusted), without re-selecting, scored PR-AUC 0.5013 on the same test partition, a lift of +0.0334. This is a textbook small-validation-set overfitting case — 27 candidate combinations compared against only 205 validation rows — not evidence that model re-selection is unsound as a general practice.

Separately from the length-matched resampling above, the engineered pipeline was re-selected on the full dataset with `word_count` folded in as an additional candidate feature, via the same 27-combination grid used in §2.3. The same architecture won again (XGBoost + SMOTE, trusted+untrusted, now 240 features including `word_count`), and test PR-AUC rose from 0.3848 to **0.4922** — closing most of the gap to TF-IDF (0.5119). Order-robust means move in the same direction (0.3676 ± 0.0182 to 0.4831 ± 0.0257), and the two configurations' bootstrap 95% confidence intervals, [0.311, 0.463] and [0.410, 0.586], barely overlap. `word_count` is accordingly treated as a first-class engineered feature in every analysis from §2.5 onward, not only as an external reference point.

3.1.5 Interpreting the pivot

Taken together, these results motivated a change in the project's central question. Comment length is associated with delta-award status more strongly than any individual engineered feature or feature family tried, and much of the apparent predictive advantage of both the original engineered pipeline and a plain bag-of-words model is attributable to this association, not to rhetorical content specifically. This is an empirical, associative finding: it establishes that length is informative and that engineered features alone did not capture that information well, not that length *causes* persuasive success, nor that every feature correlated with length is redundant with it. The length-controlled comparisons in §3.1.4 show that a modest amount of non-length signal survives in every model tested, which is the empirical basis for asking, in §3.2, whether an explicitly defined set of rhetorical strategies can characterize that signal more directly than engineered linguistic features did.

3.2 RQ2: Incremental Value and Importance of Rhetorical Strategies

3.2.1 Eight-strategy scoring and the evaluated population

Eight rhetorical strategies — justification, concession, evidence, refutation, personal narrative, call-to-action, simplification, and question-asking — were scored on an ordinal 1–5 scale by

an LLM (Gemini), following the general methodology of Labruna et al. (2026) adapted to a taxonomy defined independently for this project. Scoring was restricted to comments of at least 30 words, or of any length if delta-awarded, yielding a scored population of **2,753 comments across 400 original posts**. Because this restriction disproportionately removes short, non-convincing comments while retaining every convincing comment, the scored population's convincing-comment prevalence (641 of 2,753, 23.3%) is substantially higher than the full dataset's 16.1%; percentages reported from this point onward use this subset's own denominator unless stated otherwise.

Two reliability checks, both explicitly lightweight and not intended as a formal inter-annotator-agreement study, accompany these scores. A test-retest check (200 comments scored twice) found 95–99% of repeat scores within one point of the original across all eight strategies (mean absolute difference 0.14–0.41). A comparison against a single human judge's hand-scoring of 18 comments (9 convincing, 9 not), after the judge reviewed her own first-pass disagreements with the LLM, found at least 83.3% agreement within one point for every strategy (mean absolute difference 0.28–0.89).

Table 10. Reliability figures for each strategy (definitions in Methods, Table 5, §2.6), test-retest (n = 200) and human agreement (n = 18 — the judge's final scores after reviewing her own first-pass disagreements with the LLM), both at the same ±1-point tolerance (`05_rhetorical_scoring.ipynb`).

Strategy	Test-retest: within ±1 pt	Test-retest: mean abs. diff.	Human agreement: within ±1 pt	Human agreement: mean abs. diff.
Justification	98.5%	0.29	88.9%	0.89
Concession	99.0%	0.29	94.4%	0.67
Evidence	99.5%	0.24	88.9%	0.72
Refutation	95.5%	0.41	83.3%	0.89
Personal narrative	99.5%	0.14	94.4%	0.28
Call-to-action	95.5%	0.30	100.0%	0.67
Simplification	98.0%	0.27	100.0%	0.28
Question-asking	98.0%	0.16	94.4%	0.28

Refutation and justification are the two strategies with the weakest human-judge agreement (lowest within-±1 rates and, tied, the highest mean absolute difference of the eight). As shown below, they are also the two strategies most central to this project's predictive and interpretive findings — a pairing that qualifies, without invalidating, the results that follow.

3.2.2 Engineered baseline versus baseline plus strategy scores

The eight strategy scores were appended to a 229-feature engineered baseline (the 228-feature trusted representation with `word_count` added) and evaluated with the fixed XGBoost + SMOTE architecture, restricted to the 2,753-comment scored population (thread-grouped

split: 2,161 train / 592 test). Because of this pipeline's column-order sensitivity — reordering the same 237 columns swung PR-AUC by as much as 0.0482 in a direct check (§2.3) — every PR-AUC below is an order-robust mean ± SD across eight column orderings, not a single-order point estimate.

Table 11. Strategy-score ablation on the 2,753-comment scored population (test-fold no-skill PR-AUC 0.2382; `06_strategy_ablation.ipynb`).

Feature set	PR-AUC (mean ± SD)	ROC-AUC (mean)
Engineered baseline (no strategy scores), 229 columns	0.4434 ± 0.0187	0.7041
Engineered baseline + 8 strategy scores, 237 columns	0.5098 ± 0.0200	0.7657
Strategy scores only, 8 columns	0.3514 ± 0.0065	0.6733

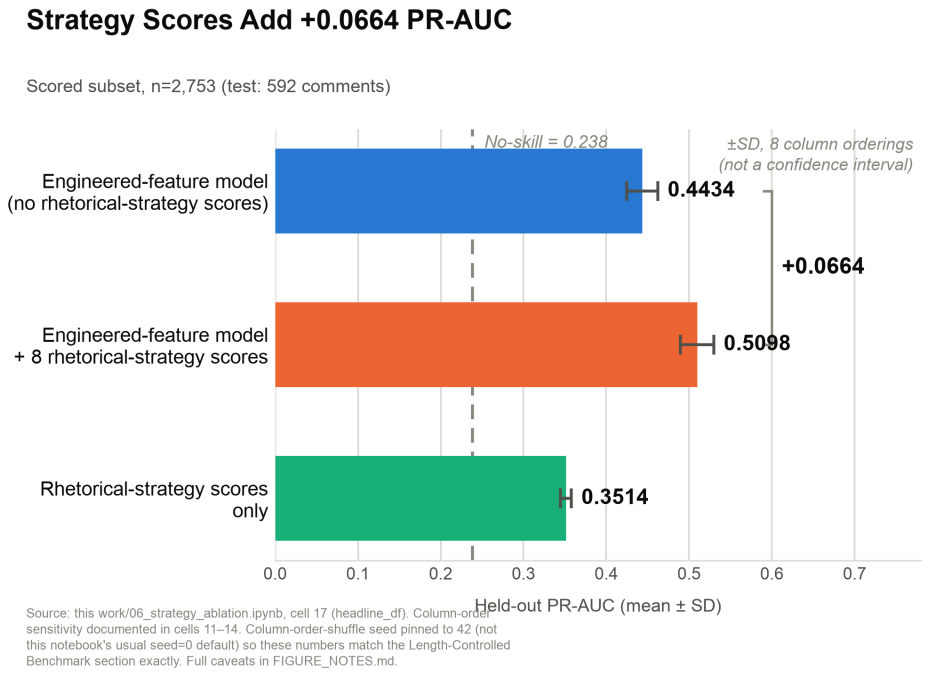


Figure 6. Held-out PR-AUC for the engineered baseline with and without the eight strategy scores, and for the strategy scores alone, on the 2,753-comment scored population (`06_strategy_ablation.ipynb`).

Adding the eight strategy scores raises PR-AUC from 0.4434 to 0.5098, a gain of **0.0664**, which clears the combined order-noise band of the two configurations. The strategy scores alone reach 0.3514 — a real, standalone signal, positioned between the scored population's own no-skill baseline (0.2382) and the combined model, but not a replacement for the existing engineered features.

3.2.3 Survival of the gain under length matching

Whether this gain reflects something beyond comment length, and not just another proxy for it, was tested on a strict decile-bin length-matched sample built within the 2,753-comment

scored population itself — independently constructed from §3.1.4's full-dataset sample of the same size (1,235 comments).

Table 12. The strategy-score gain, full scored population versus the length-matched sample built within it (`06_strategy_ablation.ipynb`).

Sample	n	Lift without strategy scores	Lift with strategy scores	Strategy-score gain
Full scored population	2,753	0.2052	0.2716	0.0664
Length-matched (within scored population)	1,235	0.0629	0.1186	0.0557

This gain is plotted in Figure 4 (§2.7), alongside the retention percentage discussed below.

Both arms' overall lift collapses sharply once length is matched (69% and 56% reductions, respectively), consistent with the pattern already established in §3.1.4: length is still doing most of the work either way. But the strategy-score *gain itself* — the increment the eight scores add on top of the engineered baseline, holding everything else fixed — drops far less steeply, from 0.0664 to 0.0557, an **84% retention**. This is a markedly smaller erosion than `word_count`'s own lift or the engineered pipeline's own lift, both of which retained only about 18% of their full-dataset values under the (differently constructed) strict match in §3.1.4. As with every length-matching result in this report, this indicates that the strategy-score signal is not primarily a proxy for comment length under the specific matching procedure applied; it does not by itself establish complete independence from length or any causal claim about the strategies' effect on persuasion.

3.2.4 Compact models

The nine-column combination and the four-column robust core reported below were identified by different routes, described fully in §2.7–§2.8: the nine-column combination by an exploratory search evaluated directly against this population's own held-out test partition, the robust core by a fixed Mann-Whitney significance criterion rather than a search. Both qualifications carry through to how the figures below should be read.

Building up from the strategy scores alone, family by family (mirroring the engineered-feature ablation methodology of §3.1), identified a configuration smaller than the full 237-column set that exceeds it: the eight strategy scores together with `word_count` alone, nine columns in total, reached order-robust PR-AUC **0.5379 ± 0.0109**, ahead of the full 237-column combination (0.5098 ± 0.0200). A second, independent estimate of this same nine-column configuration, computed in the course of the pruning check described below, gave a consistent 0.5380 ± 0.0108 on the identical test fold — a noise-level difference between the two estimates, itself illustrating the column-order spread discussed throughout this section. (A four-column subset of this same model, reported below, reaches a nominally higher PR-AUC still, but not distinguishably so; the two should be read together, not as an unqualified ranking.) Re-running the classifier/resampling grid restricted to the nine columns found Random Forest with oversampling scoring comparably to the originally selected XGBoost configuration (0.5432 ± 0.0065 versus 0.5380 ± 0.0108) — a difference smaller than the combined order-noise band of the two

architectures compared, and not a basis for preferring one over the other on this feature set; XGBoost was retained for consistency with the rest of the project.

The robust core — refutation, justification, simplification, and `word_count` (selection criterion in §2.8) — reached order-robust PR-AUC **0.5489 ± 0.0046**. Every tested removal of the other five strategies moved PR-AUC by at most −0.016, smaller than the combined order-noise band of the configurations being compared (the sum of their two respective SDs), so none of these differences is itself a confirmed effect. This is a higher point estimate than either of the nine-column estimates above, and in that sense the highest raw PR-AUC among the configurations evaluated against this population, but not distinguishably better than the nine-column model once the two configurations' combined order-noise band (0.0046 + 0.0108 = 0.0154, against a gap of only 0.0109) is taken into account. The honest reading is that question-asking, personal narrative, and evidence can be dropped without a measurable cost to predictive power, not that the four-column model is confirmed superior. The robust core was fixed in advance of the external validation reported in §3.3, not selected by searching for the best-performing subset on that external dataset.

3.2.5 Individual strategy importance: SHAP versus Mann-Whitney

Two complementary interpretive analyses were applied to the nine-column model (eight strategy scores plus `word_count`): a SHAP (SHapley Additive exPlanations) analysis using a tree explainer on the fitted XGBoost model, which describes how much the *fitted model* relies on each feature, and a model-free Mann-Whitney U comparison converted to a rank-biserial correlation effect size per strategy, with Benjamini-Hochberg false-discovery-rate correction across the eight simultaneous tests. Both were computed on the full 2,753-comment scored population and on the strict length-matched sample from §3.2.3. These two measures answer different questions — model reliance versus a model-independent univariate association — and neither is a causal-effect estimate.

Table 13. SHAP importance and Mann-Whitney rank-biserial effect size, full population versus length-matched sample, for all eight strategies (`word_count`'s own SHAP value — 1.0245 full population, 0.8402 matched — is omitted here as it dwarfs every strategy's and is not itself one of the eight strategies under comparison). * marks FDR < 0.05 significance (`07_strategy_interpretation.ipynb`).

Strategy	SHAP, full population	SHAP, length-matched	Rank-biserial r, full population	Rank-biserial r, length-matched
Justification	0.4156	0.1956	0.3584*	0.1309*
Concession	0.3439	0.3445	0.1481*	0.0206
Evidence	0.3592	0.3090	0.1018*	−0.0322
Refutation	0.6034	0.5193	0.3149*	0.1540*
Personal narrative	0.1853	0.2168	−0.0217	−0.0859*
Call-to-action	0.1837	0.2707	0.1584*	0.0592
Simplification	0.3193	0.3383	0.1944*	0.0789*
Question-asking	0.1518	0.1806	0.0069	−0.0345

The SHAP columns of this table are plotted in Figure 5 (§2.8).

On the full population, refutation has the highest SHAP importance among the eight strategies (0.6034, roughly 1.5× the next-highest), while justification has the largest rank-biserial effect size (0.3584), ahead of refutation's (0.3149). SHAP and the univariate comparison disagree about which single strategy is most important even before length is controlled for; this is a genuine point of disagreement between two legitimately different measures, not resolved by preferring one over the other.

3.2.6 Refutation, justification, and the effect of length matching

Refutation and justification diverge sharply once length is controlled for. Refutation's SHAP importance declines only modestly (0.6034 to 0.5193, holding the top rank among strategies) and its rank-biserial effect size retains 49% of its full-population value (0.3149 to 0.1540, remaining significant). Justification's SHAP importance collapses from the model's third most influential input (`word_count` and refutation ranking first and second) to one of its least influential strategies (0.4156 to 0.1956) — most of what the fitted model was using justification for on the full population appears length-related — while its rank-biserial effect size retains only 37% of its value (0.3584 to 0.1309) but *remains* statistically significant after FDR correction. Simplification is a modest, consistent third strategy across both measures (SHAP essentially unchanged, 0.3193 to 0.3383; rank-biserial retaining 41%, 0.1944 to 0.0789).

One further result is worth flagging explicitly rather than leaving implicit: personal narrative's rank-biserial effect is not significant on the full population but *is* significant, and negative, on the length-matched sample (r = −0.0859) — the only strategy whose significance status and sign both shift between the two populations. Concession, call-to-action, and evidence all lose significance entirely once length is matched, consistent with their apparent full-population effect being substantially a length artifact.

These results narrow §3.2.2–§3.2.3′s group-level finding — that strategy scores as a block carry signal beyond length — to a more specific pattern: on this dataset, that signal concentrates most clearly in refutation, with justification and simplification contributing more modestly and unevenly depending on which measure is consulted, and the remaining four strategies

contributing little that survives length control on their own. As noted in §3.2.1, refutation and justification are also the two strategies with the weakest human-judge agreement in the scoring validation — a caveat that bears directly on how strongly this specific pattern should be read, not a separate, unrelated limitation.

3.2.7 Scoring reliability and column-order sensitivity: qualifications

Two distinct sources of measurement uncertainty apply to every result in this section and should not be conflated. First, the LLM-assigned strategy scores themselves carry the reliability profile reported in §3.2.1 — strong self-consistency, weaker and more variable agreement with a single human judge on a small (n = 18) sample, with refutation and justification specifically flagged as the hardest strategies to calibrate. Second, the fitted model's PR-AUC carries the column-order sensitivity already described (§2.3, §3.2.2) — a different and independent source of uncertainty that does not reflect sampling error. Both sources bear on how confidently the refutation/ justification pattern in §3.2.6 should be stated.

Whether that specific pattern — refutation as the most length-robust strategy, justification as the most length-confounded one — reflects something general about rhetorical persuasion on CMV, or instead reflects a property of how this dataset's own non-convincing examples happen to be selected, is the question §3.3 turns to next, using an independently constructed corpus built with a different negative-example methodology.

3.3 RQ3: Cross-Corpus Evaluation on Winning Arguments

3.3.1 The WA evaluation population and evaluation tasks

The same eight-strategy rubric, scoring prompt, and fixed model architecture and training configuration already selected on the home dataset (`RobustScaler`, SMOTE oversampling, XGBoost) were applied to Winning Arguments (WA; Tan et al., 2016), an independently constructed r/changemyview corpus whose non-winning replies are deliberately selected to be topically similar to the corresponding winning reply, in contrast to the home dataset's unfiltered leaf-comment negatives (§2.1, §2.9). A total of 8,106 direct top-level replies across 3,051 original posts were collected and scored; 10 comments were dropped after an unparseable LLM response, leaving **8,096 scored comments**, evaluated on WA's own native training (6,584 comments) and held-out (1,512 comments) partitions — the same split the corpus is distributed with and on which its own published benchmark figures are computed. The held-out fold's own positive rate is 52.9%, giving a no-skill PR-AUC baseline of **0.5291** — far higher than the home dataset's no-skill baselines (roughly 0.20–0.24 depending on the fold), reflecting WA's near-balanced, pairwise construction; raw PR-AUC values are not comparable between the two datasets without reference to each one's own no-skill baseline.

The home-selected architecture and training configuration were reused on WA, but refit on WA's own native training split and evaluated on its native held-out split — a transfer of the modeling procedure, not a zero-shot evaluation of the model already fitted on the home dataset (full protocol in §2.9). Because WA was scored only on the eight-strategy rubric and `word_count`, the feature set available here is limited to combinations of the nine columns identified on the home dataset (§3.2.4), not the larger 237-column representation.

3.3.2 Predictive performance: word count, strategy scores, and the robust core

Table 14. Order-robust PR-AUC and PR-AUC lift over WA's own no-skill baseline (0.5291), for four feature-set configurations evaluated on WA's native held-out split (`08_wa_external_validation.ipynb`).

Feature set	PR-AUC (mean ± SD)	Lift over no-skill
`word_count` only	0.5767 ± 0.0000	0.0476
Strategy scores only	0.5967 ± 0.0028	0.0676
Strategy scores + `word_count`	0.5814 ± 0.0058	0.0523
Robust core (refutation, justification, simplification, `word_count`)	0.5919 ± 0.0026	0.0628

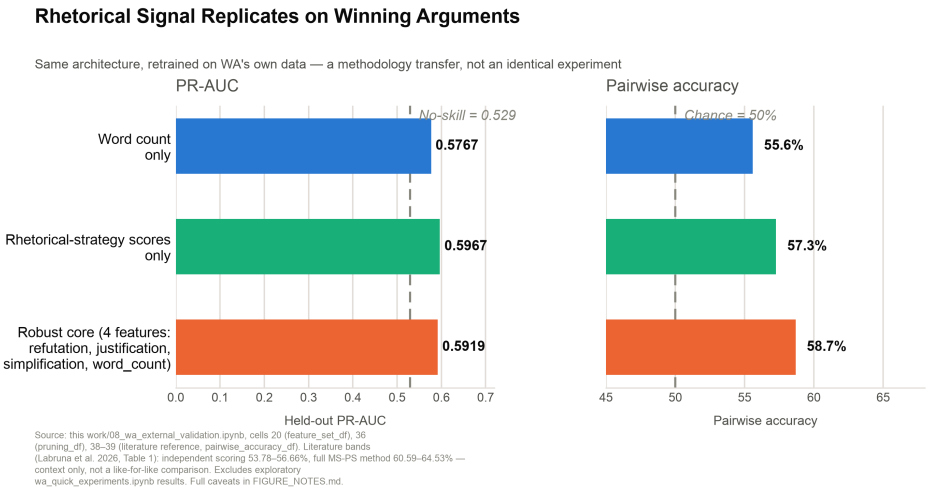


Figure 7. Held-out PR-AUC and pairwise accuracy for feature-set configurations evaluated on Winning Arguments, with literature benchmark ranges shown for context (`08_wa_external_validation.ipynb`).

Every strategy-containing configuration beats `word_count` alone by PR-AUC. Unlike on the home dataset, strategy scores alone (0.5967) slightly edge out strategy scores combined with `word_count` (0.5814) on this metric — `word_count`'s own correlation with the outcome is markedly weaker on WA (r = 0.1516) than on the home dataset (r = 0.3716), so combining it with the strategy scores has less to add here and, by this particular metric, even costs a little instead of adding.

A length-matched robustness check specific to WA — built with the same decile-matching procedure used in §3.2.3, applied to WA's own (considerably weaker) length confound — found `word_count`'s own correlation with the outcome dropping from r = 0.1516 to r = 0.1168 (a 23% reduction, much smaller than the home dataset's 79% reduction under the analogous match, because WA's winners and losers are already fairly similar in length before any matching). Comparing `word_count` alone against strategy scores plus `word_count` on this matched sample (7,710 comments), `word_count` alone lost more than half its lift (0.0476 to 0.0211, 44% retention) while the strategy-inclusive combination retained most of its lift (0.0523

to 0.0455, 87% retention) — the same qualitative pattern observed on the home dataset in §3.2.3. This check is narrower than the §3.2.2–§3.2.3 ablation on the home dataset, however: it compares `word_count` alone against `word_count` plus strategy scores, not an engineered-feature pipeline against that same pipeline plus strategy scores, since WA was never scored on the home dataset's engineered-feature pipeline. It should be read as supporting evidence that the strategy signal is not solely a length proxy on WA as well, not as a replication of the home dataset's specific engineered-pipeline-based ablation.

3.3.3 Pairwise accuracy

Pairwise accuracy — the fraction of held-out pairs for which the model scores the actual winning reply higher than the actual losing reply — was also computed, since WA's winner/loser pairs are the unit the published literature on this corpus reports accuracy against. It covers the 805 of 807 native held-out pairs with both sides successfully scored (2 pairs lost both sides to the unparseable-response drop).

Table 15. Pairwise accuracy on WA's 805 scored held-out pairs, with published literature reference ranges for context (Labruna et al., 2026, Table 1).

Feature set	Pairwise accuracy
`word_count` only	55.59%
Strategy scores only	57.27%
Strategy scores + `word_count`	58.09%
Robust core (refutation, justification, simplification, `word_count`)	58.68%
Independent Scoring (literature)	53.78%–56.66%
Multi-Strategy Persuasion Scoring (literature)	60.59%–64.53%

Within Labruna et al.'s own benchmark, "Independent Scoring" is their weakest baseline and is not itself based on the six persuasion strategies at all: the LLM is given a message alone and asked for a single holistic 1–10 persuasiveness score, with no strategy decomposition, no original-post context, and no reasoning step. "Multi-Strategy Persuasion Scoring" (MS-PS) is their actual proposed method, and is the methodologically comparable point of reference for this project's own approach: it scores each of six persuasion strategies independently by an LLM, then either averages the six scores (MS-PS-AVG) or feeds them, together with summary statistics, into a trained classifier (MS-PS-MLP) — structurally the closer analogue to this project's own strategy-scores-plus- trained-classifier pipeline. Each reported range reflects variation across the underlying LLM used to generate the scores, not a methodological difference within this project's own pipeline. The robust core's 58.68% sits above Labruna et al.'s non-strategy-based Independent Scoring baselines but below their own strategy-based MS-PS method — the comparison that matters most, given the structural similarity, favors their approach, not this project's. This is a reasonable, honestly bounded result, not a like-for-like comparison: the robust core uses none of this project's engineered features, no tuning against WA, and a different (and much simpler) scoring and modeling pipeline than the literature's own method.

3.3.4 Cross-corpus changes in strategy rankings

Table 16. SHAP importance and Mann-Whitney rank-biserial effect size for all eight strategies, home dataset versus WA, each on its own full population (`07_strategy_interpretation.ipynb`; `08_wa_external_validation.ipynb`). The two datasets' models are fit independently; SHAP magnitudes are not compared across datasets, only each dataset's own internal ranking.

Strategy	SHAP rank, home	SHAP rank, WA	Rank-biserial r, home	Rank-biserial r, WA
Refutation	1st (0.6034)	7th (0.0904)	0.3149 (2nd)	0.0949 (4th)
Justification	2nd (0.4156)	1st (0.2798)	0.3584 (1st)	0.1962 (1st)
Evidence	3rd (0.3592)	2nd (0.2136)	0.1018 (6th)	0.1467 (2nd)
Concession	4th (0.3439)	4th (0.1138)	0.1481 (5th)	0.0542 (5th)
Simplification	5th (0.3193)	3rd (0.1554)	0.1944 (3rd)	0.1105 (3rd)
Personal narrative	6th (0.1853)	8th (0.0861)	−0.0217 (8th)	0.0224 (7th)
Call-to-action	7th (0.1837)	5th (0.1107)	0.1584 (4th)	0.0305 (6th)
Question-asking	8th (0.1518)	6th (0.0953)	0.0069 (7th)	−0.0135 (8th)

(Ranks are each strategy's position among the eight strategies only, on its own dataset and measure — `word_count`'s own SHAP value, excluded from this ranking for the reasons given in §3.2.5, is 1.0245 at home and 0.2780 on WA, ahead of every strategy in both cases.)

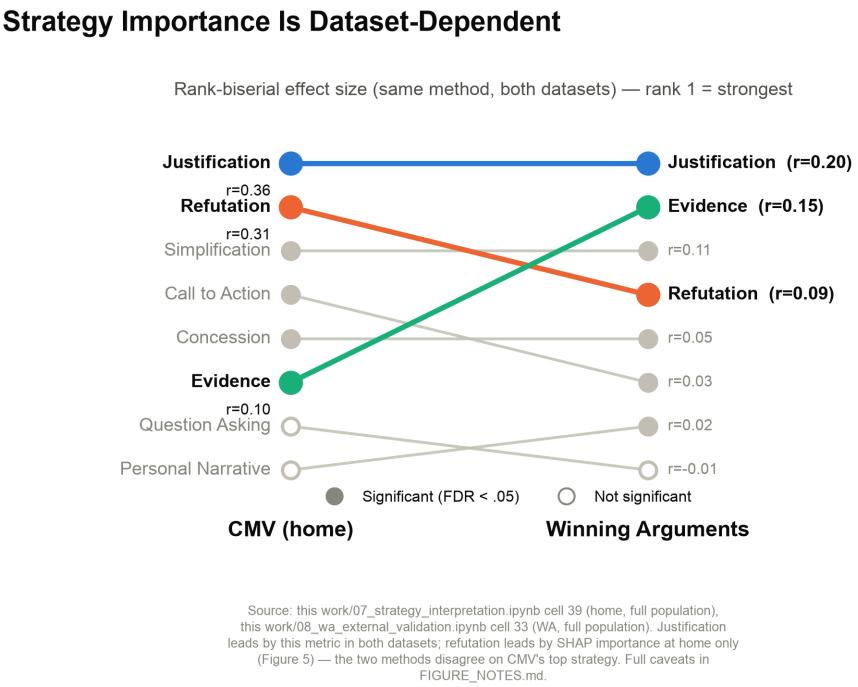


Figure 8. Rank-biserial effect size for each strategy, home dataset versus Winning Arguments, computed identically on both (`07_strategy_interpretation.ipynb`; `08_wa_external_validation.ipynb`).

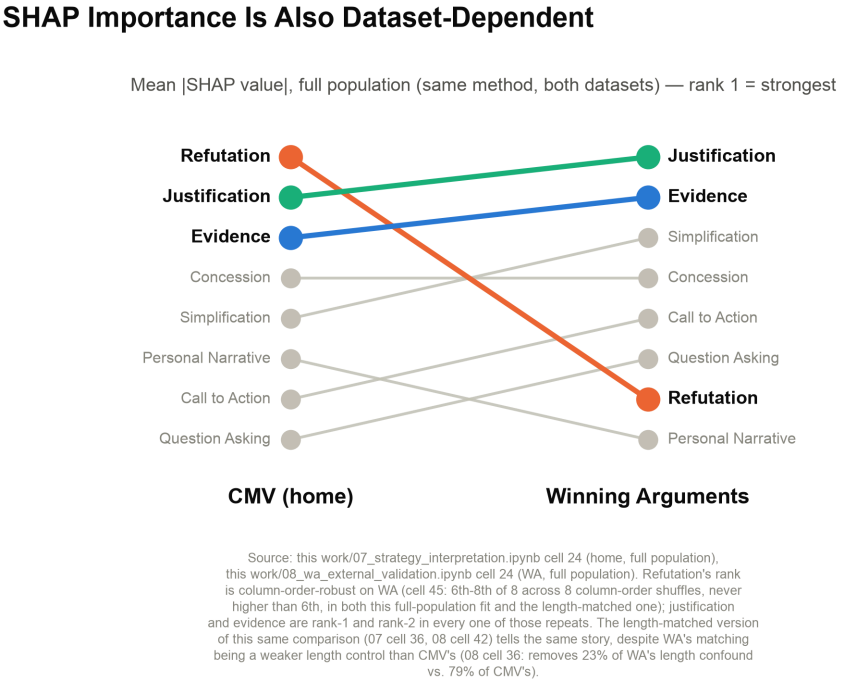


Figure 9. Mean |SHAP| *rank* (not magnitude) for each strategy, home dataset versus Winning Arguments (`07_strategy_interpretation.ipynb`; `08_wa_external_validation.ipynb`).

Refutation, the home dataset's standout strategy by SHAP, is near the bottom of WA's ranking by the same measure (7th of 8) and 4th by rank-biserial effect size. Justification, which collapsed under length control on the home dataset (§3.2.6), is WA's top strategy by both SHAP

and rank-biserial effect size, and — unlike on the home dataset — does not collapse under WA's own length-matched check: its rank-biserial effect size retains 84% of its value (0.1962 to 0.1648) rather than the 37% retained on the home dataset. A column-order-shuffle check on the WA SHAP computation (eight repeats, mirroring the home-dataset procedure) confirms refutation's low standing is stable rather than an ordering artifact: it never ranks higher than 6th of 8 strategies, in either WA's full population or its length-matched sample, across all eight repeats.

A natural question is whether this reversal reflects refutation being scored less reliably on WA. A dedicated test-retest check (n = 200, matching §3.2.1′s design) found the opposite: refutation is *more* internally consistent on WA (76.5% exact match) than on the home dataset (63.8%, the weakest of the eight strategies there), and every strategy scores more consistently on WA than on the home dataset. This test-retest check, together with the column-order-shuffle check above, finds no straightforward evidence that scoring instability explains the ranking reversal; it does not, however, rule out other, more systematic measurement differences, such as the LLM interpreting the same rubric somewhat differently against WA's text distribution than against the home dataset's, which this check is not designed to detect.

3.3.5 Pre-specified features versus configurations explored on WA

The four-feature robust core (refutation, justification, simplification, `word_count`) was fixed in advance of this external evaluation — though not before the home-corpus statistical analysis that identified it (§3.2.4, §3.2.5) — by a criterion established there: the three strategies whose Mann-Whitney effect size remained statistically significant both before and after length matching, not a search for the best-scoring feature subset. It is not the single best-performing configuration found on WA: a narrower, eight-column configuration dropping only call-to-action (identified by searching WA's own pruning results, not by any criterion fixed in advance) reached a slightly higher PR-AUC (0.5937 versus the robust core's 0.5919) and pairwise accuracy (59.10% versus 58.68%). This gap is comparable in size to this pipeline's own order-induced noise band. The robust core remains the reported external-validation result specifically because it was specified before WA was examined, not selected by scanning WA's own results for the best-performing subset; the "–call_to_action" configuration is reported here for completeness, as a configuration explored on WA, not as a replacement for the pre-specified core.

3.3.6 Summary

Taken as a whole, this external evaluation provides supporting evidence, on an independently constructed second dataset, for two structural claims: that rhetorical-strategy scores carry predictive signal beyond `word_count`, and that a small, pre-specified strategy-based feature set is competitive with, or exceeds, every larger combination tried. As §3.3.2 notes, the length-matched check behind the first of these claims is narrower in scope than the home dataset's own ablation and operates on a corpus whose own length confound is considerably weaker to begin with; the appropriate reading is a qualitatively similar pattern recurring on WA, not a replication of equal strength to the home dataset's own result. The specific claim that refutation is the leading strategy, and that justification is specifically the one whose importance is length-confounded, does not recur here — on WA, the two strategies' roles are substantially reversed. A difference between the two datasets' negative-example construction (§2.9) is a plausible candidate explanation for this reversal, but no analysis in this project tests

it directly; it is reported here as an open question and examined further, as an explicitly labeled hypothesis, in the Discussion chapter.

4. Discussion

4.1 RQ1 and RQ2: What predicts persuasive success, and which strategies carry signal beyond length

Comment length is the single strongest predictor of delta-award status on the home dataset. Both a plain bag-of-words baseline and `word_count` alone outperform the full engineered-feature pipeline (§3.1.2), and much of the apparent predictive advantage of the engineered features is attributable to their association with length, not to rhetorical content specifically (§3.1.5). This is an associative, not causal, finding: it establishes that length is informative and that the engineered features did not capture that information as well as length itself does, not that length causes persuasive success.

A real signal survives rigorous length control nonetheless (§3.1.4), and this is the empirical basis for RQ2. Adding the eight rhetorical-strategy scores to the engineered baseline produces a real, order-robust gain (0.4434 to 0.5098 PR-AUC, +0.0664, §3.2.2) that retains 84% of its value under strict length matching (0.0664 to 0.0557, §3.2.3) — markedly more robust to length control than either `word_count`'s own lift or the full engineered pipeline's lift, both of which retain only about 18% of their full-dataset values under the analogous match (§3.1.4). This is the project's central empirical conclusion for RQ2: an explicitly defined rhetorical-strategy rubric captures something beyond comment length that conventional engineered linguistic features, on their own, did not.

Resolving the SHAP/rank-biserial disagreement

Within that block-level finding, two interpretive measures disagree about which single strategy is "most important," even before length is controlled for: SHAP ranks `refutation` first (0.6034), while the rank-biserial effect size ranks `justification` first (0.3584, ahead of `refutation`'s 0.3149) (§3.2.5). This disagreement is not a contradiction to resolve by choosing a winner — it reflects two measures answering genuinely different questions, and the distinction is treated here as a conclusion in its own right, not left open.

The rank-biserial effect size is a model-free measure of marginal association: it asks how strongly a single strategy's score differs between convincing and non-convincing comments, considered in isolation from every other feature. SHAP importance is a model-dependent measure of predictive attribution: it asks how much the *fitted classifier* actually relies on a given feature once every other input — including features correlated with it — is already present in the model. A feature can score highly on one measure without scoring highly on the other precisely because the two measures are not estimating the same quantity, and disagreement between them does not indicate that either computation is wrong.

Concretely, `refutation` receives the highest SHAP attribution among the eight strategies in this specific fitted XGBoost pipeline, once `word_count` and the other seven strategies are already in the feature matrix. This describes how this particular model's output decomposes, not that `refutation` supplies predictive information absent from every other feature: with several correlated strategies present in the same feature matrix, SHAP attribution can concentrate on one of them without establishing that the others are redundant with it, or that `refutation` is the only route to the information it captures. `Justification`'s rank-biserial advantage is a different and more direct quantity: considered as an isolated pairwise relationship, `justification`'s scores covary with delta-award status more strongly than any

other strategy's scores do on their own. The two findings are compatible without being in tension — a feature's attribution within one fitted model and its own marginal association with the outcome are different quantities that need not move together, particularly once correlated predictors are present. On this basis, this report does not collapse the full-population analysis into a single "most important strategy." `Refutation` is the strongest model-attributed feature in this specific fitted pipeline; `justification` is the strategy whose score differs most, on its own, between convincing and non-convincing comments. Neither claim should be read as a general or causal statement about which rhetorical strategy is most persuasive on CMV, and the two should not be merged into one ranking.

The two strategies diverge sharply once length is controlled for, and this divergence is informative independent of which baseline measure one prefers. `Refutation` retains the large majority of both measures under length matching — SHAP declines only modestly (0.6034 to 0.5193, holding the top rank among strategies) and the rank-biserial effect size retains 49% of its value (0.3149 to 0.1540, remaining significant). `Justification`'s SHAP importance, by contrast, collapses from the model's third most influential input to one of its least influential strategies (0.4156 to 0.1956) — most of what the fitted model was using `justification` for on the full population appears length-related — while its rank-biserial effect size retains only 37% of its value (0.3584 to 0.1309) but remains statistically significant after FDR correction (§3.2.6). The conclusion drawn here is therefore specific, not global: `refutation` is the home dataset's most length-robust rhetorical strategy by *both* interpretive measures, even though the two measures disagree about which strategy is most important before length is taken into account.

The `personal_narrative` sign flip

One further pattern deserves comment, not just a bare mention. `Personal_narrative`'s rank-biserial effect is not statistically significant on the full population, but is significant and *negative* on the length-matched sample ($r = -0.0859$, §3.2.6) — the only strategy whose significance status and sign both shift between the two populations. One plausible interpretation is that length is masking a genuine negative association in the full population: a tendency for heavy personal-narrative use to co-occur with lower persuasive success may be offset, on average, by a correlated and opposing length effect, becoming visible only once length variation is reduced. This is offered here explicitly as one candidate interpretation, not an established explanation. The length-matched sample is a separately and specifically constructed subset of the data (§2.5, §3.2.3), not simply a length-corrected version of the same population with every other property held fixed, so the comparison does not isolate length as the only variable that changed between the two analyses. An equally available alternative is that the full-population effect is simply not robust and the length-matched result is closer to noise. Distinguishing between these honestly would require further analysis beyond this project's scope; the result is reported here as a pattern worth noting, not as a resolved finding.

4.2 RQ3: Cross-corpus evaluation — what recurs, what does not, and why that matters

The external evaluation on Winning Arguments (WA) provides supporting evidence, on an independently constructed corpus built with a different negative-sampling methodology, for two structural claims: that rhetorical-strategy scores carry predictive signal beyond

word_count, and that a small, pre-specified strategy-based feature set is competitive with, or exceeds, larger combinations (§3.3.2, §3.3.5, §3.3.6).

This evidence deserves substantial weight as independent-corpus support for the central RQ2 claim, though it is best read as a qualitatively similar pattern recurring on a second corpus rather than a replication of equal strength to the home dataset's own result (see the scope and length-confound caveats below). WA's own length-matched robustness check (§3.3.2) found the strategy-inclusive combination retaining 87% of its lift over no-skill (0.0523 to 0.0455) while word_count alone retained only 44% (0.0476 to 0.0211) — the same qualitative asymmetry already established on the home dataset (84% versus roughly 18%, §3.2.3, §3.1.4). That a second corpus, built by a different team with a deliberately different sampling procedure, shows the same qualitative asymmetry is meaningful supporting evidence that this project's central claim — rhetorical-strategy scores are not simply a proxy for comment length — is not an artifact specific to how this project's own dataset happens to be constructed.

That said, the strength of this conclusion should be calibrated to what the check actually measures. WA's own length confound is considerably weaker to begin with: matching reduces word_count's own correlation with the outcome by only 23% on WA, against a 79% reduction under the home dataset's analogous match (§3.3.2) — WA's winning and losing replies are already fairly similar in length before any matching is applied, since the corpus's own sampling is topic-matched instead of unfiltered. The WA check is therefore a somewhat less demanding test than the one the home dataset's own ablation poses, and it compares a narrower pair of configurations (word_count alone against word_count plus strategy scores), not the home dataset's full engineered-pipeline-against- itself-plus-strategies comparison, since WA was never scored on that pipeline (§2.9, §3.3.2). The appropriate conclusion is that the strategy signal recurs qualitatively on a second corpus under a narrower and easier test, not that its magnitude or its independence from length has been established to the same degree as on the home dataset.

The specific claim that refutation leads that signal, and that justification is specifically the strategy whose importance is length-confounded, does not recur. On WA, the two strategies' relative roles are substantially reversed: refutation falls from first to seventh of eight by SHAP and from second to fourth by rank-biserial effect size, while justification rises to first by SHAP and remains first by rank-biserial in both datasets — and, unlike at home, justification's rank-biserial effect does *not* collapse under WA's own length matching, retaining 84% of its value (0.1962 to 0.1648) against the 37% retained at home (§3.3.4). These checks do not find straightforward evidence that scoring instability explains the reversal: a dedicated test-retest check found refutation scored *more* consistently on WA (76.5% exact match) than on the home dataset (63.8%, the weakest of the eight strategies there), and a column-order-shuffle check confirms refutation's low WA standing is stable across eight repeats, never ranking above sixth of eight (§3.3.4). This rules out instability in the scoring process itself as the explanation, but not other, more systematic measurement differences — for instance, the LLM interpreting the same rubric somewhat differently against WA's text distribution than against the home dataset's — which neither check is designed to detect.

A plausible, but explicitly untested, explanation is that the two corpora's negative-example construction differs in a way that matters for which strategy looks most informative. The home dataset's non-convincing comments are the thread's own unfiltered population; WA's

non-winning replies are deliberately selected to be topically similar to the winning reply within the same thread (§2.9, §3.3.1). If `refutation`'s apparent importance at home partly reflects its ability to distinguish a persuasive reply from topically *unrelated* or off-target replies — a distinction WA's sampling procedure specifically removes by construction — this could account for its collapse on WA without requiring any change in how persuasion itself works. This is offered as one candidate hypothesis consistent with the observed pattern, not as a tested or established cause; no analysis in this project isolates the sampling-methodology variable directly, and other explanations (e.g., differences in topic distribution, thread structure, or the populations' own base rates) have not been ruled out either.

Taken together, this points to a specific, bounded conclusion for RQ3: the general claim that rhetorical-strategy scores add predictive information beyond length generalizes across two independently constructed CMV-adjacent corpora; the specific claim that one particular strategy drives that signal does not generalize, and may instead depend on properties of how a dataset's negative examples are selected. That dependency is itself a substantive empirical observation about the limits of strategy-level generalization in this research area, not merely a limitation of this project's own results.

4.3 Methodological insights

A recurring pattern across this project is worth naming explicitly as a learned concept, not just as background to the final numbers. Three times, an initial result was taken seriously enough to be tested against itself and, in each case, revised before being reported as found: the discovery that `word_count` alone outperforms the full engineered pipeline, which prompted the length-controlled analysis in the first place (§2.4, §3.1); a small-validation-set overfitting failure caught when model selection was blindly re-run on the length-matched sample, where the re-selected architecture scored *below* its own no-skill baseline on test while the originally selected architecture continued to score positively (§2.5); and the discovery that this project's XGBoost+SMOTE pipeline is sensitive to the order of columns in a correlated, 200+-column feature matrix, large enough (a 0.0482 PR-AUC swing in a direct check, §2.3) to rival some of the effects under study. Each of these was disclosed and corrected, not absorbed silently, and the third prompted an explicit methodological convention adopted mid-project: every PR-AUC reported from that point onward *for an evaluation of this order-sensitive XGBoost+SMOTE architecture* is a mean ± SD across multiple column orderings, not a single-order point estimate — this does not apply to the logistic-regression-based baselines (TF-IDF, `word_count` alone) or the dummy classifiers compared against it, which have no column-order degree of freedom and continue to be reported as single-fit point estimates, typically alongside a bootstrap confidence interval instead (§2.3). The resulting uncertainty discipline — reporting column-order means and SDs rather than single-fit point estimates for this architecture — carries through every later comparison in this report and is reflected in the limitations stated in §4.4, below.

It is also worth stating plainly where this project's deep-learning and transformer-based techniques actually sit in the pipeline, since the final predictive architecture is not itself a neural model: the selected classifier throughout is XGBoost with SMOTE oversampling, a gradient-boosted tree ensemble. Deep learning is used instead as a feature-generation and representation-learning layer upstream of that classifier. This includes an LLM (Gemini), used both for several engineered features (sentiment, tone, style, and the ethos/pathos/logos rhetor-

ical-appeal scores, §2.2) and, centrally, for the entire eight-strategy rhetorical rubric that is this project's main methodological contribution (§2.6); transformer-based sentence embeddings (`BAAI/bge-base-en-v1.5`, reduced by PCA to roughly 187 components at ≥85% variance, §2.2); two HuggingFace transformer classifiers that were evaluated and explicitly rejected after qualitative inspection — a RoBERTa emotion classifier, two-thirds of whose predictions collapsed to "neutral," and a zero-shot `bart-large-mnli` classifier with visible mislabeling on spot checks — in favor of LLM scoring (§2.2); and one HuggingFace transformer-based persuasive-language detector (`chreh/persuasive_language_detector`) that was kept as a feature. Reporting the two rejected models alongside the two kept ones is deliberate: the choice of which deep-learning tool to use for a given feature was itself evaluated empirically, not assumed in advance.

4.4 Limitations

- The LLM-scored rhetorical strategies carry a reliability ceiling that is not uniform across strategies, and it is concentrated on exactly the two strategies central to §4.1′s headline interpretation: `refutation` and `justification` have the weakest human-judge agreement of the eight strategies scored (83.3% and 88.9% within one point respectively, against 94.4–100% for most others, §3.2.1). This does not invalidate the SHAP/rank-biserial pattern discussed above, but it bounds how strongly that pattern should be stated — it is possible that some part of what distinguishes these two strategies reflects scoring noise and not a stable underlying difference.
- Most individual point-estimate comparisons in this project carry overlapping uncertainty bands, and none was accompanied by a dedicated statistical test of the difference itself (e.g., a paired bootstrap or permutation test) — they should be read as this project's best point estimates, not as confirmed or statistically significant rankings. Two comparisons stand out as the largest, most consistent margins found anywhere in this project, and are the ones this report leans on most heavily: the original `word_count`-versus-engineered-pipeline discovery (§3.1.2, §2.4), where the two configurations' bootstrap 95% confidence intervals — [0.311, 0.463] without `word_count`, [0.410, 0.586] with it — barely overlap rather than failing to overlap outright; and the strategy-ablation gain (§3.2.2), where the two configurations' PR-AUC differ by more than their combined column-order- noise bands, which is evidence against an essentially arbitrary implementation detail (column order), not a sampling-based significance test (§2.3). Neither comparison is described here as statistically significant or decisive in the absence of a dedicated test of the difference; both are reported as the most persuasive point-estimate gaps this project found, nothing stronger. The nine-column-versus-robust-core comparison (§3.2.4) illustrates the more typical case: the robust core's higher point estimate is not distinguishable from the nine-column model once their combined order-noise band is taken into account.
- A second, independent source of uncertainty layers on top of the first: the fitted model's PR-AUC is itself sensitive to the order of the (correlated, 200+-column) feature matrix's columns, a property unrelated to sampling error (§2.3, §2.7, §3.2.7). Both sources bear on how confidently any single reported number in this report should be read.
- The WA external evaluation supports the *group-level* claim that strategy scores carry signal beyond length, not a claim of equal strength to the home dataset's own length-controlled ablation: the WA length-matched check is narrower in scope (§3.3.2, §4.2 above) and operates on a corpus whose own length confound is considerably weaker to begin with.

- The dataset is drawn from a single subreddit, a single platform, and is English-only; thread size is also structurally correlated with a thread's apparent convincing rate (one delta per thread roughly fixed, while negative examples scale with thread branching), which is relevant to any analysis that aggregates at the level of the original post instead of the individual comment (§2.1).
- Delta-award status is an operational proxy for persuasive success, not direct evidence of a lasting change in belief (§1.1). Every interpretive claim in this chapter concerns what predicts this observable, incentive-shaped label, not persuasion as a psychological construct in general.

4.5 Possible improvements and future work

The following are proposed directions for future work, not analyses already carried out in this project:

- Test the negative-example-construction hypothesis from §4.2 directly, for instance by constructing a topic-matched negative-sampling variant of the home dataset itself, or by stratifying WA's own replies by topical similarity to the winning reply and checking whether `refutation`'s standing shifts systematically with that property. This would isolate whether the sampling-methodology difference, specifically, drives the cross-corpus reversal, rather than some other difference between the two corpora.
- Extend external validation to a third, differently constructed persuasion corpus. A third dataset would help establish whether the structural claim (RQ2′s core finding) generalizes beyond two corpora, and whether its own strategy ranking aligns more closely with the home dataset's or with WA's — informative either way, given the reversal already observed between the two corpora evaluated here.
- Conduct a more rigorous, multi-judge inter-annotator-agreement study of the rhetorical-strategy rubric, particularly for `refutation` and `justification`. The reliability check performed in this project is explicitly a lightweight check against a single judge on a small sample (§3.2.1), and a larger, multi-judge study would help establish whether these two strategies' weaker LLM-human agreement reflects genuine ambiguity in the underlying construct or is better attributed to the limited power of this project's own check.

4.6 Conclusions

Returning to the three research questions stated in the Introduction:

RQ1. Comment length is the dominant single predictor of delta-award status on the home dataset. Conventional engineered textual features, evaluated on their own, add little predictive power beyond what length already explains, and a naive comparison that does not account for length risks attributing predictive power to rhetorical content that is substantially a length effect instead.

RQ2. An explicitly defined, eight-strategy rhetorical rubric captures real predictive signal beyond comment length that this project's conventional engineered features did not. That signal is not uniform across strategies: `refutation` is the strategy whose contribution is most robust to length control by both of two legitimately different interpretive measures, while `justification` carries the largest marginal association with the outcome overall but loses most of its distinct predictive attribution once length is accounted for. These two measures

— model-based attribution and model-free marginal association — answer different questions and are not reducible to a single ranking.

RQ3. The structural claim from RQ2 — that rhetorical-strategy scores add information beyond length — finds supporting evidence on an independently constructed corpus built with a different sampling methodology, under a narrower and somewhat easier test than the home dataset's own ablation. The specific claim about which strategy drives that signal does not recur: `refutation` and `justification`'s relative importance substantially reverses on Winning Arguments; the project's reliability checks find no straightforward evidence that scoring instability explains this, though they do not rule out other, more systematic measurement differences between the two corpora. A plausible, but explicitly untested, candidate explanation is that the two corpora's differing negative-example construction changes what a given strategy is effectively being asked to distinguish.

Across all three questions, this project's contribution is primarily empirical and analytical: a documented account of how strongly comment length predicts persuasive success and why that matters for interpreting any persuasion-prediction model; a systematic, length-controlled assessment of a purpose-built rhetorical-strategy rubric's incremental value, including which specific strategies drive that value under which measure; and an external evaluation that identifies both what recurs and what does not when the same approach is applied to an independently constructed dataset. The project does not claim a new learning algorithm, and its findings about individual strategies are associative and measure-dependent rather than causal — both limitations stated here deliberately, not left implicit.

5. References

5.1 Cited literature

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5.2 Code reuse

The Reddit data-collection approach (§2.1) and the evidence-marker keyword-stem feature (§2.2) both adapt code released by Priniski and Horne alongside their 2018 study — `collect_posts.py` and `discussion_analysis.py`, respectively — rather than reimplementing either independently. The code is available at https://github.com/jpriniski/CMV.

5.3 Systems and libraries

The pipeline draws on PRAW (Reddit API client), Google's `google-genai` SDK (Gemini), HuggingFace Transformers models (a persuasive-language detector, `chreh/persuasive_language_detector`, and a sentence-embedding model, `BAAI/bge-base-en-v1.5`, both retained; a RoBERTa-based emotion classifier and a zero-shot `bart-large-mnli` classifier, both trialed and rejected per §2.2), spaCy, `textstat`, XGBoost, `imbalanced-learn` (SMOTE), SHAP, and ConvoKit (used to access the Winning Arguments corpus for external validation, §2.9).

5.4 AI tools used

Google Gemini was used as the LLM underlying several engineered features (tone, formality, and rhetorical-appeal scores) and the rhetorical-strategy scoring rubric described in §2.2–§2.6; its role in the research method is documented throughout the Methods chapter. Claude Code (Anthropic) was used as an AI coding assistant during the development of the data-processing and modeling pipeline and the accompanying notebooks, and also assisted with drafting and editing this report's text. The research design, analysis, and conclusions are the author's own.

5.5 Code and Data Availability

The project's notebooks, supporting code (`common_functions.py`), and `README.md` are included directly with this submission. `README.md` documents setup, the notebook execution order, and the project's working data files. The raw Reddit data is not included in the submission due to its size; it is available via the read-only Google Drive link given in `README.md`. The Winning Arguments corpus used for external validation (§2.9) is fetched automatically through ConvoKit and is not distributed as part of this submission.

Appendix A

Table 2: Feature-Family Summary

Table 2, referenced from §2.2, is reproduced here in full because its five columns do not fit comfortably within a single portrait page. It is split into three parts (Table 2a–2c) so that each fits cleanly on one page.

Table 2a. Feature-family summary, part 1 of 3 — sentiment, tone, and style families. With each family's source notebook and whether it belongs to the trusted or untrusted set. For the two one-hot-encoded rows here, the category count in the feature-group name is the true number of labels the classifier assigns; "Columns" is one less per side, since the first category of each encoding is dropped as an implicit reference level (§2.2).

Feature / feature group	Variants included	Source notebook(s)	Columns	Set
Sentiment (LLM, tone-focused)	comment, OP, diff	03.1, 03.3	3	Trusted
Sentiment (VADER, lexicon-based)	comment only	03.1	1	Untrusted
Tone (LLM, 6 categories, one-hot)	comment, OP	03.1, 03.3	10 (2 sides × 5, first category dropped per side)	Trusted
Tone (RoBERTa emotion classifier, 7 categories, one-hot)	comment only	03.1	6 (1 side × 6, first category dropped)	Untrusted
Style / formality (LLM-assigned, binary)	comment, OP	03.1, 03.3	2	Trusted
Style (BART-large-MNLI zero-shot, 2 categories, one-hot)	comment only	03.1	1 (1 side × 1, first category dropped)	Untrusted

Table 2b. Feature-family summary, part 2 of 3 — rhetorical appeals, style/readability, and relational features.

Feature / feature group	Variants included	Source notebook(s)	Columns	Set
Rhetorical appeals: ethos, pathos, logos (LLM, 0–1 scale)	comment, OP, diff	03.2, 03.3	9	Trusted
Rhetorical appeals: ethos, pathos, logos (Empath lexicon)	comment only	03.2	3	Untrusted
Adjective–adverb ratio	comment, OP, diff	03.2, 03.3	3	Trusted
Readability: Flesch–Kincaid grade level	comment, OP, diff	03.2, 03.3	3	Trusted
Readability: Flesch reading ease	comment only (no OP/diff counterpart — dropped as redundant with `fk_grade_level_op`, r = −0.905)	03.2	1	Trusted
Evidence-marker count	comment, OP, diff	03.2, 03.3	3	Trusted
Comment↔OP match indicators (sentiment, tone, style, ethos, pathos, logos)	boolean agreement, computed from original comment/OP values, not derived from the "diff" columns above (bucketed same/different, not a thresholded difference)	03.3	6	Trusted

Table 2c. Feature-family summary, part 3 of 3 — the two remaining feature families, plus the representation totals carried forward into model selection (§2.3).

Feature / feature group	Variants included	Source notebook(s)	Columns	Set
Persuasive-language detector output	comment only	03.2	1	Trusted
Document embeddings (`BAAI/bge-base-en-v1.5`, PCA)	comment only	03.2	187	Trusted
Handcrafted Trusted Total			**41**	Trusted
Handcrafted Untrusted Total			**11**	Untrusted
Trusted (handcrafted + embeddings)			**228**	—
Trusted + untrusted			**239**	—